Wood Shell Tongue Drum Capstone

Instrument-Maker Print Packet

Build packet folder: wood-shell-tongue-drum

Generated: 2026-05-08

This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

File Map

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
photo-shotlist.md	Project artifact.
risks.md	Project artifact.
validation-report.md	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Wood Shell Tongue Drum — Parametric Design Sheet

Owner: Tony Koop · Heifer Zephyr Instruments

Family: Slit-tongue idiophone with enclosed wooden resonator (round-body)

Members: four geometric variants (V1–V4) × three size envelopes (Travel 12″, Standard 16″, Floor Pouf 20″)

Status: Design v1 — formulas parametric in `wood-shell-tongue-drum-design-table.xlsx`; no physical prototype yet

Recommended first prototype: **V1 Cylinder + Flat Top, Standard 16″, A Kurd**

---

1. Intent

This is the round-body counterpart to Tony's existing rectangular-prism wood tongue drum. A wooden shell — cylinder or hemisphere — capped by a tonewood soundboard with CNC-cut slit tongues, tuned to a chosen scale, is a modern instrument category that does not exist commercially in this round-body × dome-top × premium-tonewood format. The enclosed cavity adds **Helmholtz resonance** that an open rectangular tongue drum cannot offer — warm bass coupling beneath the lowest tongue.

The sheet treats this as a *family with four variants* rather than as four separate designs. All twelve cells (4 variants × 3 sizes) are driven by the same shared inputs: shell OD, shell height, soundboard thickness/wood, gu-port diameter, dome rise. Change one preset cell in the workbook and every dimension in the matrix updates.

ID	Body	Top	Notes
V1	Cylinder	Flat	Lowest piece-count, most predictable tuning. **Start here.**
V2	Cylinder	Domed	Steel-tongue-drum aesthetic on a wooden body
V3	Hemisphere bowl	Flat	Tagine / Moroccan-pouf silhouette; bowl as parabolic reflector
V4	Hemisphere bowl	Domed	Most steel-tongue-drum-adjacent silhouette in wood; near-spherical

Size	OD (in)	Height (in)	Soundboard (in)	Tongue width (in)	Target ding	Ding f (Hz)
Travel	12	5	0.250	1.00	D4 (MIDI 62)	293.66
Standard	16	8	0.375	1.25	A3 (MIDI 57)	220.00
Floor Pouf	20	10	0.500	1.50	D3 (MIDI 50)	146.83

(All values in this document are computed from the workbook formulas, not pasted by hand. If you change the workbook, regenerate the BOM and validation CSVs.)

---

2. Governing acoustic model

The wood shell tongue drum is a **two-degree-of-freedom coupled oscillator**: the cantilever tongue is one resonator, the enclosed air cavity is another, and they share a common boundary (the soundboard).

2.1 Tongue — flat cantilever (Euler–Bernoulli)

For a slit tongue of length `L` (in) and thickness `t` (in) in wood with cantilever K-constant `K` (imperial composite):

$f = K \cdot t / L^2$ (flat top, Variants 1 and 3)

$L = \sqrt{(K \cdot t / f)}$

Tony's material library (workbook §A, rows 19–25) records measured / literature K-constants for shop tonewoods:

Wood	E (GPa)	ρ (kg/m^3)	K (cantilever)	Tone	Rating
Padauk	11.0	745	24,438	warm, rich, sustain	★★★★★
Wenge	15.8	870	27,103	bright, punchy	★★★★★
Cherry	10.3	560	27,275	balanced sweet mid	★★★★
Black Walnut	11.6	610	27,734	full, mellow	★★★★
Hard Maple	12.6	705	26,887	bright sustain	★★★★
Mahogany	10.1	590	26,314	warm, full bass	★★★★
W. Red Cedar	7.7	370	29,013	light, guitar-like	★★★

Padauk K = 24,438 is the lowest in the library and therefore yields the longest tongues for a given pitch — the design defaults to Padauk because it is most permissive of fitting low pitches in a given shell radius.

2.2 Tongue — curved cantilever (V2, V4)

A tongue cut into a domed soundboard is stiffer than a flat tongue. To preserve pitch we apply an empirical length correction:

$L_{curved} = L_{flat} \cdot 1.025$ (+2.5 % for shallow domes, dome rise ≤ 6 % of OD)

This is a **starting estimate**, not a measured constant. The dominant uncertainty in V2 and V4 is whether 1.025 holds for our specific dome rise. Validation protocol: tap-tune one tongue at the design length, measure with a tuner, back-solve the actual factor, then apply to the remaining ten tongues. Treat the +2.5 % multiplier as *the empirical-learning loop's first measurement*.

2.3 Cavity — Helmholtz resonator

For an enclosed chamber of volume `V` (in^3) with port area `A_port` (in^2) and effective neck length `L_neck` (in):

$f_H = (c / 2\pi) \cdot \sqrt{A_{port} / (V \cdot L_{neck})}$ with $c = 13{,}552$ in/s

Chamber volume formulas, by variant:

Variant	Body shape	Cavity volume formula
V1	Cylinder, flat top	$V = \pi \cdot r^2 \cdot H$
V2	Cylinder + dome	$V = \pi \cdot r^2 \cdot H + (2/3) \cdot \pi \cdot r^2 \cdot h_{dome}$
V3	Hemisphere, flat	$V = (2/3) \cdot \pi \cdot r^3$
V4	Hemisphere + dome	$V = (2/3) \cdot \pi \cdot r^3 + (2/3) \cdot \pi \cdot r^2 \cdot h_{dome}$

Effective neck length:

Top type	`L_neck`
Flat	soundboard thickness `t`
Domed	`t + h_dome / 2`

The **Helmholtz target** is `f_H ≤ f_ding` — the cavity should resonate at or below the lowest tongue, so it acts as a bass extension rather than a competing pitch. Coupling falls into one of three regimes:

- `f_H / f_ding < 0.8` — free bass extension; cavity reinforces below the ding (good for floor pouf).
- `0.8 ≤ f_H / f_ding ≤ 1.2` — coupled regime; cavity and ding share modal energy (the design target).
- `f_H / f_ding > 1.2` — cavity sits above the ding; cavity will fight the ding rather than support it. **Mistuned high** — lower `f_H` by reducing effective port area, adding a removable restrictor/liner, lengthening the neck, or increasing the chamber volume.

The **gu port diameter** is the primary tuning knob. Because `f_H ∝ sqrt(A_port)`, opening the port raises the Helmholtz frequency; reducing effective area or increasing neck length lowers it. The port is drilled last, deliberately undersized, and opened up incrementally with a step bit while measuring `f_H` (gentle puff at the port, microphone above). This is the same field-tuning method used on Hapi-style steel tongue drums.

2.4 Computed Helmholtz / ding ratios (current presets)

Each cell below is `f_H` (Hz) computed from the formulas above with the current size-preset inputs, and the ratio against `f_ding` for that size. Bold cells sit in the coupled regime (0.8 ≤ ratio ≤ 1.2) — that is the buildable matrix.

Variant	Travel 12″ (D4 ding 293.66 Hz)	Standard 16″ (A3 ding 220.00 Hz)	Floor Pouf 20″ (D3 ding 146.83 Hz)
V1	321.5 Hz · ratio **1.09** (tight)	**194.6 Hz · ratio 0.88**	**144.7 Hz · ratio 0.99**
V2	220.1 Hz · ratio 0.75	133.5 Hz · ratio 0.61 (under-coup)	99.1 Hz · ratio 0.67 (under-coup)
V3	359.5 Hz · ratio 1.22 (over)	**238.3 Hz · ratio 1.08**	177.2 Hz · ratio 1.21 (over, edge)
V4	244.2 Hz · ratio 0.83	161.1 Hz · ratio 0.73	119.5 Hz · ratio 0.81

Reading the matrix: **V1 Standard 16″ and V1 Floor Pouf 20″ are in the sweet spot**; V3 Standard 16″ is also coupled. V2 cells are systematically under-coupled because the dome adds neck length without proportional volume — they will need a *larger* effective gu port than the preset 2.5″, a shorter effective neck, or a revised dome cavity model to lift `f_H`. V1 Travel 12″ is right at the upper edge of the coupled band.

2.4.1 Gu-port tuning direction

Condition	Field symptom	Corrective direction	First shop action
`f_H/f_ding < 0.80`	Cavity too low / under-coupled	Raise `f_H`	Enlarge gu port in 0.25″ steps; if already at structural max, reduce neck length or revise chamber volume
`0.80 ≤ f_H/f_ding ≤ 1.20`	Coupled	Hold	Stop drilling; record final port diameter and environment
`f_H/f_ding > 1.20`	Cavity too high	Lower `f_H`	Install removable restrictor/liner, lengthen neck with a port sleeve, or increase chamber volume on the next build

For the Standard 16″ V2 preset, the 2.5″ port predicts 133.5 Hz (`0.61 × ding`). A first-order diameter solve puts the lower coupled edge near 3.3″ and the `0.95 × ding` target near 3.9″, so the V2/V4 validation build should treat the port as a measured design variable, not a fixed decorative hole.

These values are **derived estimates**, not measured. They will shift once the first prototype is tuned and the per-family corrections database is populated.

2.5 Tongue-fit cap

For each size, the longest tongue (the ding) cannot exceed the radial fit cap:

L_cap = OD/2 − 1″ (1″ reserved for ding-strike clearance and rim)

Size	Cap (in)	V1/V3 ding length (in)	V2/V4 ding length (in)	Verdict
Travel 12″	5.00	4.56 (D4)	4.68 (D4)	Fits — 9 % margin
Standard 16″	7.00	6.45 (A3)	6.62 (A3)	Fits — 8 % margin
Floor Pouf	9.00	9.12 (D3)	9.35 (D3)	**Doesn't fit**

Floor Pouf D3 ding overhangs by 0.12″ on flat tops and 0.35″ on domed tops. **Mitigation:** raise the ding to D■3 (one semitone, MIDI 51, 155.6 Hz → flat L = 8.86″, fits with 0.14″ margin) OR thin the soundboard from 0.5″ → 0.4375″ (drops L to 8.53″, clean 0.47″ margin) OR substitute Cedar (K = 29,013 → flat L = 8.34″, 0.66″ margin). Cedar is the recommended fix because the soundboard span across 20″ wants stiffness against deflection, and Cedar's lower density lets the same thickness still bear the span.

---

3. Variant blocks — manufacturing and acoustic notes

Each block extends the workbook's §C–§F by spelling out the *manufacturing path* implied by the geometry.

V1 — Ottoman Cylinder, Flat Top ★ RECOMMENDED FIRST BUILD

Body construction: stave-built or segmented cylinder. The two paths are equally valid:

• **Stave shell**: 16 staves at 11.25° miter (180/16). Rip from 4/4 stock, joint edges, band-clamp glue-up around an MDF cylinder mold. Lathe-true OD/ID after cure (≥ 24 h). Wall thickness after turning: 5/8″–3/4″.

• **Segmented shell** (used when polychrome inlay is desired): 8 rings × 12 segments at 15° miter. Stack rings on a lazy-susan jig for glue-up. Diamond pattern with Walnut + Maple + Cherry accents reads beautifully on the Ottoman 16″ size.

Top construction: flat tonewood disc, CNC-cut to OD and edge-rabbeted ¼″ deep into the shell rim. Slits routed *before* glue-up so the tongues vibrate without router stress. Disc diameter = shell OD; thickness from preset.

Tongue physics: standard flat cantilever. Empirical detuning ≤ 1 % observed on Tony's prior rectangular drums — trust the model.

Risk / Difficulty: ★★ — proven flat-cantilever physics + well-trodden stave/segmented bowl method. Lowest risk in the matrix.

V2 — Ottoman Cylinder, Domed Top

Body construction: same as V1.

Top construction: glue-up blank (3 plies × ~0.5″ = 1.5″ thick), flatten on jointer, then 3D-surface a parabolic dome (rise = preset 0.75″ on Standard) on the CNC. Bottom contour matches OD rim. Tongue slits cut afterward in one of two ways: (a) 1/8″ upcut spiral on a B-axis sled that tracks the dome curvature (preferred — preserves the dome face) or (b) flatten a sacrificial fixture under the dome to support tongue ends, then route slits with a regular 3-axis pass.

Tongue physics: curved cantilever stiffer than flat. The +2.5 % length correction is built into the workbook. Validate empirically: tap-tune ONE tongue and back-fit the multiplier *before* cutting all 11.

Risk / Difficulty: ★★★ — adds dome-surfacing risk and curved-cantilever uncertainty. Prototype the dome in poplar before committing Padauk.

V3 — Hemisphere Bowl, Flat Top

Body construction: segmented (ring-stack of decreasing diameters approximating a hemisphere) or thick-glue-up CNC-carved hemisphere. Either path requires lathe-true rim flatness — the bowl-to-soundboard joint is acoustically critical.

Top construction: flat soundboard disc (same as V1). The shell rim must be precisely circular and planar — turn the rim true on the lathe *after* segment glue-up. Rabbet for the board is ¼″ deep.

Tongue physics: same flat cantilever as V1. The bowl shape doesn't affect tongue physics, only the Helmholtz coupling and projection pattern (the bowl acts as a parabolic reflector, projecting tongue energy upward through the slits).

Risk / Difficulty: ★★ — bowl turning is straightforward at 12–16″; > 18″ needs a lathe with ≥ 10″ swing over bed.

V4 — Hemisphere Bowl, Domed Top

Body construction: glued bowl (segmented or stave) + glue-up dome blank, both 3D-surfaced on the CNC. Half-lap or rabbet joinery at the rim. Final assembly clamps with a custom radius caul.

Top construction: same dome workflow as V2.

Tongue physics: curved cantilever +2.5 % correction (same as V2). The flat → curved physics shift is the dominant uncertainty in V4; budget extra prototype tongues.

Risk / Difficulty: ★★★★ — highest piece-count + highest empirical tuning risk. Most beautiful result; **build last**.

---

4. Recommended first prototype — V1 Cylinder + Flat Top, Standard 16″

This is the **strongly recommended starting build**. Justification:

1. **Acoustic predictability.** Flat-cantilever physics has < 1 % empirical detuning on Tony's prior rectangular drums; the dome correction in V2/V4 is unmeasured.
2. **Helmholtz already coupled.** Standard 16″ V1 sits at `f_H / f_ding = 0.88` — inside the coupled band by design, with the gu port (2.5″) as the in-shop tuning knob if the measured ratio drifts.
3. **All 11 tongues fit cleanly.** Longest tongue (A3 ding) = 6.45″ vs the 7″ radial cap → 8 % margin. No risk of running out of soundboard.
4. **Lowest piece-count.** 16 staves + 1 disc = 17 wood pieces. Compare V4 segmented-bowl-plus-dome at 96+ pieces.
5. **Dual-path body construction.** If staves don't inspire the maker, the same shell can be built segmented for polychrome aesthetics (see V1 body construction notes).
6. **Validated workflow components.** Stave glue-up, lathe truing, CNC slit routing — all are skills exercised on Tony's existing ashiko, conga, and tongue-drum repos.

Specific prototype build configuration:

Parameter	Value
Shell OD	16.00″
Shell height	8.00″
Shell wood	Black Walnut, 4/4 stock, 16 staves at 11.25° miter
Wall thickness	0.625″ finished (after lathe truing)
Soundboard wood	Padauk (K = 24,438)
Soundboard thk	0.375″ (3/8″)
Disc diameter	16.00″ (rabbet ¼″ deep × ■″ wide into shell rim)
Scale	A Kurd (A3, B3, C4, D4, E4, F4, G4, A4, B4, C5, D5)

Ding	A3 (220.00 Hz), centered
Number of tongues	11 (1 ding + 10 in zigzag-ascending pattern)
Slit kerf width	0.125″ (1/8″ CNC upcut spiral)
Tongue width	1.25″
Gu port Ø	2.5″ initial; final size set by Helmholtz tuning
Predicted f_H	194.6 Hz (0.88 × ding)

Tongue length schedule (Padauk 0.375″, A Kurd transposed from workbook §A reference scale):

#	Note	MIDI	Freq (Hz)	L (in)	L (mm)
Ding	A3	57	220.00	6.454	163.93
1	E4	64	329.63	5.273	133.93
2	F4	65	349.23	5.123	130.12
3	G4	67	392.00	4.835	122.81
4	A4	69	440.00	4.564	115.92
5	B4	71	493.88	4.308	109.41
6	C5	72	523.25	4.185	106.30
7	D5	74	587.33	3.950	100.33
8	E5	76	659.26	3.728	94.70
9	G5	79	783.99	3.419	86.84
10	A5	81	880.00	3.227	81.97

(These are *cut lengths* of the slit, not the kerf. The CNC pocketing geometry — radial slit + perpendicular tongue-base relief — is detailed in `drawing-brief.md` and `cnc/README.md`.)

---

5. Build sequence — phases across the family

Phases are designed so each one produces a finished, playable instrument *and* a measurement that informs the next phase. This is the empirical-learning loop running across the four-variant matrix.

Phase	Build	What it teaches
1	**V1 Standard 16″** ★ HERE	Shell stave construction, soundboard rabbet joint, gu-port tuning, Helmholtz coupling — all in one go. Confirms `f_H` model against measurement.
2	V2 Standard 16″ (same shell)	Curved-cantilever +2.5 % correction. Reuse Phase-1 cylinder shell — only the soundboard changes. Quantifies the dome correction factor.

3	V3 Standard 16″	Hemisphere-bowl construction independent of dome work. Now we know flat-top + bowl independently.
4	V4 (Standard 16″ or Floor 20″)	Combine learnings. Premium tonewoods, signature Heifer Zephyr finish.
5	V1 Travel 12″	Tightest tongue-fit margins; build last when the formulas are calibrated to your shop.

This sequence delivers four distinct portfolio instruments while progressively retiring acoustic risk. Each phase's measurements feed back into `validation.csv` and the per-family corrections in the workbook.

---

6. Critical joints and tolerances

Joint	Tolerance	Why
Shell rim flatness	± 0.005″ TIR	Bowl-to-soundboard joint is the primary air seal (Helmholtz Q)
Soundboard rabbet fit	0.005″ slip-fit	Tight enough for PVA bond, loose enough to avoid prying
Stave miter angle	11.25° ± 0.1°	A 0.5° error compounds over 16 staves to a 8° total deviation
Slit kerf width	0.125″ ± 0.005″	Wider kerfs lower tongue Q (energy leaks through the kerf)
Tongue length (per cut)	+ 0.030″ initial	Cut long; file to pitch with a sanding bar (file shortens → raises pitch)
Gu port concentricity	± 0.025″	Off-center port shifts effective neck length and `f_H`
Wall thickness uniformity	± 0.030″	A wedge-shaped wall makes the cavity asymmetric — kills bass

---

7. Open questions and assumptions (red-team flagged in `risks.md`)

The design table is parametric, but the following are **derived estimates** that need first-prototype measurements to harden:

• **Padauk K = 24,438** is from the workbook's material library, derived from textbook E and ρ values. Not measured on Tony's specific stock.

• **Curved-cantilever +2.5 %** correction is a starting estimate from cylinder-shell theory at small radii. The actual factor for our specific dome rise (0.75″ on Standard) is unmeasured.

• **Wall thickness** of 0.625″ is conventional but un-tuned — Helmholtz Q depends on wall stiffness.

• **Bowl-top joint type.** Three options compete (rabbet flush, lap step, cork gasket). Rabbet flush is the working assumption; the others are validation B/C plans if the rabbet leaks.

• **Slit kerf cap of 50 % of soundboard area** is a structural rule of thumb, not a measured failure threshold.

• **Effective neck length on V1 = soundboard thickness alone.** This ignores the air "plug" oscillating just below the slit. The end correction is small for tongue-drum kerfs (kerfs are slits, not circular ports), and we're absorbing the error into the 0.88-ratio safety margin. If `f_H` measures 5–10 % low on first prototype, this is the likely cause.

---

8. Cross-references

• **Workbook**: [`wood-shell-tongue-drum-design-table.xlsx`](wood-shell-tongue-drum-design-table.xlsx) — single sheet, 249 rows, 190 formulas. Blue cells are inputs; everything else recomputes.

• **Concept sheet**: [`concepts/round-body-variants_concept-sheet.png`](concepts/round-body-variants_concept-sheet.png) — visual layout of all twelve cells.

• **BOM, sourcing, cut list, validation**: see the four CSVs at repo root.

• **Risks**: see [`risks.md`](risks.md) for the red-team pass with verification tests attached.

• **Wolfram exploration**: see [`wolfram-starter.wl`](wolfram-starter.wl) for the 3-DOF coupled oscillator (tongue + shell + air cavity) starter.

• **Sister repos**: [`tongue-drum/`](https://github.com/tonykoop/tongue-drum) (rectangular-prism precursor) · [`steel-tongue-drum/`](https://github.com/tonykoop/steel-tongue-drum) (steel cantilever cousin) · [`ceramic-tongue-drum/`](https://github.com/tonykoop/ceramic-tongue-drum) (slip-cast cousin) · [`ashiko-drum-workshop/`](https://github.com/tonykoop/ashiko-drum-workshop) (segmented bowl reference) · [`conga/`](https://github.com/tonykoop/conga) (stave & segmented shell reference).

• **Catalog**: [`tonykoop/instrument-maker`](https://github.com/tonykoop/instrument-maker) — orchestrating skill and Master Catalog.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

part_id	description	variant	size	quantity	unit	unit_cost_usd	extended_cost_usd	critical	notes
SHELL-STAVE-WALNUT-T	Black Walnut stave, 4/4 stock, 6.5 in length	V1+V2	Travel 12in	16	each	3.00	48.00	Y	Rip from 4/4; bevel 11.25 deg
SHELL-STAVE-WALNUT-S	Black Walnut stave, 4/4 stock, 9.5 in length	V1+V2	Standard 16in	16	each	4.50	72.00	Y	Rip from 4/4; bevel 11.25 deg
SHELL-STAVE-WALNUT-F	Black Walnut stave, 4/4 stock, 11.5 in length	V1+V2	Floor Pouf 20in	16	each	5.50	88.00	Y	Rip from 4/4; bevel 11.25 deg
SHELL-SEG-WALNUT-T	Black Walnut segment blank, 4/4 stock, 4 in piece	V1+V2 (alt)	Travel 12in	96	each	0.50	48.00	N	Alt body: 8 rings x 12 segment
SHELL-SEG-WALNUT-S	Black Walnut segment blank, 4/4 stock, 5 in piece	V1+V2 (alt)	Standard 16in	96	each	0.65	62.40	N	Alt body: 8 rings x 12 segment
SHELL-SEG-WALNUT-F	Black Walnut segment blank, 4/4 stock, 6 in piece	V1+V2 (alt)	Floor Pouf 20in	96	each	0.80	76.80	N	Alt body: 10 rings x 12 segmen
BOWL-SEG-WALNUT-T	Black Walnut segment blank, 4/4 stock (graduated rings)	V3+V4	Travel 12in	72	each	0.55	39.60	N	6 rings of decreasing dia for he
BOWL-SEG-WALNUT-S	Black Walnut segment blank, 4/4 stock (graduated rings)	V3+V4	Standard 16in	96	each	0.65	62.40	N	8 rings of decreasing dia for he
BOWL-SEG-WALNUT-F	Black Walnut segment blank, 4/4 stock (graduated rings)	V3+V4	Floor Pouf 20in	120	each	0.80	96.00	N	10 rings of decreasing dia for h
BOWL-BLANK-WALNUT-T	Black Walnut glue-up blank for CNC bowl, 13x13x4 in	V3+V4 (alt)	Travel 12in	1	each	85.00	85.00	N	Alt body: 4x 1in plies. CNC 3D
BOWL-BLANK-WALNUT-S	Black Walnut glue-up blank for CNC bowl, 17x17x5 in	V3+V4 (alt)	Standard 16in	1	each	140.00	140.00	N	Alt body: 5x 1in plies. CNC 3D
BOWL-BLANK-WALNUT-F	Black Walnut glue-up blank for CNC bowl, 21x21x6 in	V3+V4 (alt)	Floor Pouf 20in	1	each	210.00	210.00	N	Alt body: 6x 1in plies. CNC 3D
SOUNDBOARD-PADAUK-T	Padauk soundboard disc, 12.25 in OD, 0.25 in	V1+V3	Travel 12in	1	each	42.00	42.00	Y	CNC-cut to OD. Slits routed be
SOUNDBOARD-PADAUK-S	Padauk soundboard disc, 16.25 in OD, 0.375 in	V1+V3	Standard 16in	1	each	68.00	68.00	Y	CNC-cut to OD. Slits routed be
SOUNDBOARD-PADAUK-F	Padauk soundboard disc, 20.25 in OD, 0.5 in	V1+V3	Floor Pouf 20in	1	each	98.00	98.00	Y	CNC-cut to OD. Slits routed be
SOUNDBOARD-CEDAR-F-ALT	Western Red Cedar soundboard disc, 20.25 in OD, 0.5 in	V1+V3 alt	Floor Pouf 20in	1	each	55.00	55.00	N	Alternate to Padauk for tight D
DOMEBLANK-PADAUK-T	Padauk dome glue-up blank, 13x13 in, 1.5 in thick (3 plies x 0.5)	V2+V4	Travel 12in	1	each	75.00	75.00	Y	3D-surface 0.5 in dome rise or
DOMEBLANK-PADAUK-S	Padauk dome glue-up blank, 17x17 in, 1.5 in thick (3 plies x 0.5)	V2+V4	Standard 16in	1	each	125.00	125.00	Y	3D-surface 0.75 in dome rise c
DOMEBLANK-PADAUK-F	Padauk dome glue-up blank, 21x21 in, 1.5 in thick (3 plies x 0.5)	V2+V4	Floor Pouf 20in	1	each	180.00	180.00	Y	3D-surface 1 in dome rise on C
DOMEBLANK-POPLAR-S	Poplar dome blank for prototype validation, 17x17 in, 1.5 in	V2+V4	Standard 16in	1	each	32.00	32.00	N	Cheap test blank to dial in dom
GUE-TITEBOND3	Titebond III wood glue, 16oz	All	All	2	bottle	18.00	36.00	Y	Waterproof; 4000 psi shear. Ra
RABBET-BIT-RBR	Rabbeting router bit set with bearing kit, 1/4 in shank	V1+V3	All	1	set	42.00	42.00	N	For shell rim rabbet. One-time
SLIT-BIT-UPCUT-1-8	Solid carbide upcut spiral bit, 1/8 in, 1/4 in shank	All	All	3	each	16.00	48.00	Y	For tongue slits. Buy 3 (one br
BALLEND-3-4	Ball-end mill, 3/4 in, 1/2 in shank	V2+V4	All	1	each	38.00	38.00	Y	For dome 3D-surfacing on CN
GU-STEPBIT-3	Step drill bit, 1/4 to 3 in, 1/4 in increments	All	All	1	each	28.00	28.00	Y	For incremental gu-port enlarg
ORING-RUBBER-25	Rubber O-ring, 2.5 in ID, 1/8 in cross-section	V1+V3	Standard 16in	1	each	2.50	2.50	N	Optional gu-port lining for clea
ORING-RUBBER-30	Rubber O-ring, 3 in ID, 1/8 in cross-section	V1+V3	Floor Pouf 20in	1	each	3.00	3.00	N	Optional gu-port lining
ORING-RUBBER-20	Rubber O-ring, 2 in ID, 1/8 in cross-section	V1+V3	Travel 12in	1	each	2.00	2.00	N	Optional gu-port lining
FELT-FOOT-PADS	Adhesive felt floor protection pads, 1 in dia	All	All	4	each	0.50	2.00	N	Mount under bowl/cylinder bot
HANDLE-ROPE-S	Hemp rope, 0.5 in, for carry handle	All	Standard 16in+	4	ft	2.50	10.00	N	Pass through two 0.5 in holes
FINISH-DANISH-OIL	Watco Danish Oil, natural, qt	All	All	1	each	28.00	28.00	N	3 coats, 24h between
FINISH-WAX-CARNAUBA	Renaissance Wax, 65ml	All	All	1	each	18.00	18.00	N	Final coat on shell exterior
SANDPAPER-SET	Sandpaper assortment 80/120/220/320/400/600 grit	All	All	1	set	32.00	32.00	N	
TUNER-KORG-OT120	Korg OT-120 wide-range tuner	All	All	1	each	180.00	180.00	N	Shop tool. Already owned in T
MALLET-SET	Tongue drum mallets, soft rubber heads, pair	All	All	1	set	18.00	18.00	N	Player accessory, included wit

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

part_id	supplier	supplier_url	sku_or_search	lead_time_days	moq	verified_price_date	price_volatility	required_spec	search_terms	notes
SHELL-STAVE-WALNUT-S	Bell Forest Products	bellforestproducts.com	black walnut 4/4 S2S FAS	5	1 BdFt	2026-05-07	low	4/4 S2S FAS black walnut, edge-jointable	black walnut 4/4 turning blanks	Recommended supplier for s
SHELL-STAVE-WALNUT-T	Bell Forest Products	bellforestproducts.com	black walnut 4/4 S2S FAS	5	1 BdFt	2026-05-07	low	4/4 S2S FAS black walnut	black walnut 4/4	
SHELL-STAVE-WALNUT-F	Bell Forest Products	bellforestproducts.com	black walnut 4/4 S2S FAS	5	1 BdFt	2026-05-07	low	4/4 S2S FAS black walnut	black walnut 4/4	
SHELL-SEG-WALNUT-S	Cook Woods	cookwoods.com	black walnut 4/4 turning blanks	4	1	2026-05-07	low	Bulk segment-grade walnut	walnut turning blanks 4/4	Alt source for segmented bod
BOWL-SEG-WALNUT-S	Cook Woods	cookwoods.com	black walnut 4/4 turning blanks	4	1	2026-05-07	low	Graduated-diameter ring blanks	walnut turning blanks 4/4	
BOWL-BLANK-WALNUT-S	Cook Woods	cookwoods.com	black walnut 8/4 thick stock	10	1	2026-05-07	medium	5x 1in plies for CNC bowl glue-up	walnut 8/4 thick	CNC bowl alt path; verify thic
SOUNDBOARD-PADAUK-S	Bell Forest Products	bellforestproducts.com	padauk 3/8 figured S4S	5	1 BdFt	2026-05-07	medium	Padauk 3/8 in thickness, FAS	padauk 4/4 S4S, edge-quartersawn	Critical thickness; quartersawn preferred t
SOUNDBOARD-PADAUK-T	Bell Forest Products	bellforestproducts.com	padauk 1/4 S4S	5	1 BdFt	2026-05-07	medium	Padauk 1/4 in thickness, FAS	padauk thin stock	
SOUNDBOARD-PADAUK-F	Bell Forest Products	bellforestproducts.com	padauk 1/2 S4S	5	1 BdFt	2026-05-07	medium	Padauk 1/2 in thickness, FAS	padauk 1/2 in	
SOUNDBOARD-CEDAR-F-ALT	OldGrowth Music	oldgrowthmusic.com	western red cedar soundboard 1/2 quartersawn	7	1	2026-05-07	medium	WRC 1/2 in quartersawn, A grade	western red cedar soundboard 1/2	Alt for D3 ... Floor Po
DOMEBLANK-PADAUK-S	Bell Forest Products	bellforestproducts.com	padauk 4/4 boards (3 boards required)	5	1 BdFt	2026-05-07	medium	3x 1/2in boards, edge-jointable	padauk 4/4 S4S	Glue 3 plies to make dome b
DOMEBLANK-POPLAR-S	Lowe's	lowes.com	poplar 1x12 board (3 boards)	1	1	2026-05-07	low	3x poplar boards 1x12 in	poplar 1x12 x 8	Cheap test stock for dome su
GUE-TITEBOND3	Rockler	rockler.com	Titebond III 16oz	2	1	2026-05-07	low	PVA wood glue, waterproof, 4000 psi shear	Titebond III	
RABBET-BIT-RB	Rockler	rockler.com	rabbeting router bit set with bearing kit	2	1	2026-05-07	low	1/4 in shank, multi-bearing rabbet kit	rabbeting bit set	
SLIT-BIT-UPCUT-1-8	Amana Tool	amanatool.com	46202 1/8in upcut spiral	3	1	2026-05-07	low	Solid carbide, 1/8 in, 1/4 in shank, upcut	Amana 46202 1/8 upcut spiral	
BALLEND-3-4	Amana Tool	amanatool.com	46377 3/4 ball nose	3	1	2026-05-07	low	3/4 in ball nose, 1/2 in shank, solid carbide	Amana 46377 3/4 ball nose	
GU-STEPBIT-3	Amazon	amazon.com	Irwin Unibit step drill 1/4 to 3 in	2	1	2026-05-07	low	Step bit covering 0.25 to 3/4 in in increments	step drill bit 1/4 to 3/4 in	
ORING-RUBBER-25	McMaster-Carr	mcmaster.com	9452K42 Buna-N O-ring 2-1/2 ID 1/8 thick	2	1	2026-05-07	low	Buna-N O-ring 2.5 in ID, 1/8 in cross	O-ring 2.5 ID 1/8	
ORING-RUBBER-20	McMaster-Carr	mcmaster.com	9452K38 Buna-N O-ring 2 ID 1/8 thick	2	1	2026-05-07	low	Buna-N O-ring 2 in ID, 1/8 in cross	O-ring 2 ID 1/8	
ORING-RUBBER-30	McMaster-Carr	mcmaster.com	9452K46 Buna-N O-ring 3 ID 1/8 thick	2	1	2026-05-07	low	Buna-N O-ring 3 in ID, 1/8 in cross	O-ring 3 ID 1/8	
HANDLE-ROPE-S	Knot and Rope Supply	knotandrope.com	hemp rope 1/2 in 25 ft hank	3	1	2026-05-07	low	Natural hemp rope, 1/2 in dia	hemp rope 1/2 in	
FELT-FOOT-PADS	Hardware store	local	felt floor-protection pads	1	1	2026-05-07	low	Adhesive 1 in felt circles	felt furniture pads 1 in	
FINISH-DANISH-OIL	Rockler	rockler.com	Watco Danish Oil natural quart	2	1	2026-05-07	low	Penetrating oil finish, natural color	Watco Danish Oil natural quart	
FINISH-WAX-CARNAUBA	Rockler	rockler.com	Renaissance Wax 65ml	2	1	2026-05-07	low	Microcrystalline wax for final finish	Renaissance Wax	
SANDPAPER-SET	Hardware store	local	sandpaper assortment 80-600 grit	1	1	2026-05-07	low	Mixed grits 80/120/220/320/400/600	sandpaper assortment	
MALLET-SET	Hapi Drum	hapidrum.com	silicone tongue drum mallet pair	5	1	2026-05-07	low	Soft rubber/silicone heads, 8 in shaft	tongue drum mallet pair	

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

variant	size	operation_id	description	stock_id	qty	length_in	width_in	thickness_in	rough_dimensions	final_dimensions	total_bdft	notes
V1+V2	Standard 16in	S-STAVE-01	Stave blank rough cut from 4/4 walnut	SHELL-STAVE-WALNUT-S	16	9.5	3.5	0.875	9.5 x 3.5 x 0.875	9.0 x 3.25 x 0.875	[illegible]	Allows trim to final 8 in she
V1+V2	Travel 12in	T-STAVE-01	Stave blank rough cut from 4/4 walnut	SHELL-STAVE-WALNUT-T	16	6.5	2.7	0.875	6.5 x 2.7 x 0.875	6.0 x 2.45 x 0.875	[illegible]	Allows trim to final 5 in she
V1+V2	Floor Pouf 20in	F-STAVE-01	Stave blank rough cut from 4/4 walnut	SHELL-STAVE-WALNUT-F	16	11.5	4.4	0.875	11.5 x 4.4 x 0.875	11.0 x 4.1 x 0.875	[illegible]	Allows trim to final 10 in sh
V1+V2	All	STAVE-EDGE	Stave miter bevel both edges 11.25 deg	N/A	16	N/A	N/A	N/A	N/A	N/A	N/A	Tilting-arbor jig referenced
V1+V3	Standard 16in	S-DISC-01	Padauk soundboard disc CNC cut	SOUNDBOARD-PADAUK-S	1	16.25	16.25	0.375	16.25 x 16.25 x 0.375	16.0 dia x 0.375	0.69	Disc dia = shell OD. Rabbe
V1+V3	Travel 12in	T-DISC-01	Padauk soundboard disc CNC cut	SOUNDBOARD-PADAUK-T	1	12.25	12.25	0.25	12.25 x 12.25 x 0.25	12.0 dia x 0.25	[illegible]	
V1+V3	Floor Pouf 20in	F-DISC-01	Padauk soundboard disc CNC cut	SOUNDBOARD-PADAUK-F	1	20.25	20.25	0.5	20.25 x 20.25 x 0.5	20.0 dia x 0.5	1.41	Consider Cedar substitute
V2+V4	Standard 16in	S-DOME-BLANK	Padauk dome blank glue-up 3 plies	DOME-BLANK-PADAUK-S	3	17	17	0.5	17 x 17 x 1.5 (after glue-up)	17 x 17 x 1.5	[illegible]	Glue 3 plies. Cure 24 h. Fl
V2+V4	Standard 16in	S-DOME-CNC	3D-surface dome rise 0.75 in	N/A	1	N/A	N/A	N/A	17x17 blank	16.0 dia x 1.5 max thickness, 0.75 in dome rise	N/A	3/4 in ball-end, 0.05 in step
V2+V4	Standard 16in	S-DOME-CNC-PROTO	3D-surface dome on poplar test blank first	DOME-BLANK-POPLAR-S	1	17	17	1.5	17x17x1.5	16.0 dia, 0.75 in dome rise	N/A	De-risk dome surfacing on
V3+V4	Standard 16in	S-BOWL-RING-01	Hemisphere bowl ring 1 (largest, foot)	BOWL-SEG-WALNUT-S	12	2.5	1.7	0.875	2.5 x 1.7 x 0.875	stack height 0.75	N/A	15 deg miter, 12 segments
V3+V4	Standard 16in	S-BOWL-RING-02	Hemisphere bowl ring 2	BOWL-SEG-WALNUT-S	12	2.4	1.7	0.875	2.4 x 1.7 x 0.875	stack height 0.75 cumulative 1.5	N/A	Outer dia ~14.7 in
V3+V4	Standard 16in	S-BOWL-RING-03	Hemisphere bowl ring 3	BOWL-SEG-WALNUT-S	12	2.3	1.7	0.875	2.3 x 1.7 x 0.875	stack height 0.75 cumulative 2.25	N/A	Outer dia ~13.0 in
V3+V4	Standard 16in	S-BOWL-RING-04	Hemisphere bowl ring 4	BOWL-SEG-WALNUT-S	12	2.0	1.7	0.875	2.0 x 1.7 x 0.875	stack height 0.75 cumulative 3.0	N/A	Outer dia ~11.0 in
V3+V4	Standard 16in	S-BOWL-RING-05	Hemisphere bowl ring 5	BOWL-SEG-WALNUT-S	12	1.7	1.7	0.875	1.7 x 1.7 x 0.875	stack height 0.75 cumulative 3.75	N/A	Outer dia ~9.0 in
V3+V4	Standard 16in	S-BOWL-RING-06	Hemisphere bowl ring 6	BOWL-SEG-WALNUT-S	12	1.4	1.4	0.875	1.4 x 1.4 x 0.875	stack height 0.75 cumulative 4.5	N/A	Outer dia ~7.0 in
V3+V4	Standard 16in	S-BOWL-RING-07	Hemisphere bowl ring 7	BOWL-SEG-WALNUT-S	12	1.1	1.1	0.875	1.1 x 1.1 x 0.875	stack height 0.75 cumulative 5.25	N/A	Outer dia ~5.0 in
V3+V4	Standard 16in	S-BOWL-RING-08	Hemisphere bowl ring 8 (apex)	BOWL-SEG-WALNUT-S	12	0.8	0.8	0.875	0.8 x 0.8 x 0.875	stack height 0.75 cumulative 6.0	N/A	Outer dia ~3.0 in. Cap with
V3+V4	Standard 16in	S-BOWL-BASE	Hemisphere bowl base disc (alternative apex)	BOWL-SEG-WALNUT-S	1	3	3	0.875	3 x 3 x 0.875	2.75 dia x 0.875	N/A	Substitutes for ring 8 if des
V1+V2	Standard 16in	S-SHELL-LATHE	Lathe true OD/ID after stave glue-up	N/A	1	N/A	N/A	N/A	OD 16.5 ID 14.5	OD 16.0 ID 14.75	N/A	Wall thickness 0.625 in fini
V1+V3	Standard 16in	S-SLITS-CNC	CNC tongue slits in soundboard	N/A	1	N/A	N/A	N/A	see drawing-brief	11 slits per radial layout, 1/8 kerf	N/A	Cut from inside face of disc
V2+V4	Standard 16in	S-SLITS-DOME	CNC tongue slits in domed soundboard	N/A	1	N/A	N/A	N/A	see drawing-brief	11 slits, 1/8 kerf	N/A	Slits cut AFTER 3D surfac
All	All	GU-PORT	Drill gu port through bowl/shell bottom	GU-STEPBIT-S	1	N/A	N/A	N/A	start 0.25 in pilot	step up 0.25 in increments to 2.0/2.5/3.0 (per size)	N/A	Tune Helmholtz f_H to ~0.
All	All	SAND	Progressive sand 80-400 grit	SANDPAPER-SET		N/A	N/A	N/A	N/A	N/A	N/A	Inside and outside. Critical
All	All	FINISH-OIL	Apply Watco Danish Oil	FINISH-DANISH-OIL	3	N/A	N/A	N/A	N/A	N/A	N/A	3 coats, 24 h between. Sou
All	Standard 16in	HANDLE-DRILL	Drill handle holes	N/A	2	N/A	N/A	N/A	0.5 in dia	0.5 in through holes, 4 in apart	N/A	Optional. Skip on Travel si

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Drawing Brief — Wood Shell Tongue Drum

The `drawings/` folder ships dimensioned SVGs for the matrix. Each SVG follows Heifer Zephyr's standard title block (lower-right) with: drawing name, sheet number, scale, units, project (Wood Shell Tongue Drum), date, revision, drafter (Tony Koop). All dimensions in inches unless noted; metric equivalents in brackets where helpful.

Required drawings

File	Sheet	Title	Notes
`01-variant-matrix.svg`	1/9	Variant Matrix V1–V4 × Three Sizes (top-down + side profiles)	Twelve-cell layout matching `concepts/round-body-variants_concept-sheet.png` with overlaid critical dims
`02-v1-standard-side-section.svg`	2/9	V1 Standard 16″ — Side Section A-A	Recommended first prototype. Shell, soundboard, rabbet joint, cavity, gu port
`03-v1-standard-tongue-layout.svg`	3/9	V1 Standard 16″ — Tongue Field (A Kurd 11-note)	Top-down soundboard view. Slit centerlines, lengths, position numbering
`04-shell-stave-template.svg`	4/9	Stave Template — All Sizes (full-size cutting layout)	One stave per size; bevel callouts at 11.25°; centerline marked
`05-shell-segment-template.svg`	5/9	Segment Template — All Sizes (alt body construction)	15° miter, 12-per-ring; 8 rings on Std, 6 on Travel, 10 on Floor
`06-rabbet-joint-detail.svg`	6/9	Bowl-Top Rabbet Joint — Section B-B (3× scale)	¼″ × ■″ rabbet; soundboard slip-fit; glue line; air-seal callout
`07-dome-toolpath.svg`	7/9	V2/V4 Dome Surfacing — CNC Toolpath Schematic	Convex top + concave bottom; 0.05″ stepover; flip-jig datum pins
`08-bowl-segment-stack.svg`	8/9	V3/V4 Hemisphere Bowl — Ring Stack Section	8 rings of decreasing dia; lathe-true profile overlay
`09-gu-port-tuning-progression.svg`	9/9	Gu Port Tuning — Step-Drill Progression	0.25″ pilot → step increments → final size; f_H expected at each step

Drawing conventions

- **Datums:** shell bottom plane = primary horizontal datum (Z=0); axis of revolution = primary vertical datum.
- **Tolerances:** ± 0.020″ default; tighter where called out (rim flatness ± 0.005″, slit kerf width ± 0.005″, miter angle ± 0.1°).
- **Cross-section hatching:** oak-grain pattern at 45° for wood; cross-hatch at 90° for void/cavity.
- **Title block size:** 4″ × 1.5″ at 1:1 scale.

• **Critical dimensions:** bold, with leader lines pointing to the dimensioned feature. Helmholtz-driving dimensions (cavity volume parameters) are double-boxed.

• **Material color cues** (when polychrome): Walnut = warm brown fill, Maple = pale cream, Cherry = pink-brown, Padauk = orange-red. Used in sheet 5 segment-stack diagrams for the diamond pattern.

Critical dimensions (must appear on the drawings)

Dimension	Drawing	Value (Standard 16″ V1)	Tolerance
Shell OD	02	16.000″	± 0.020″
Shell ID	02	14.750″	± 0.030″
Shell wall thickness	02	0.625″	± 0.030″
Shell height	02	8.000″	± 0.030″
Soundboard OD	02, 03	16.000″	± 0.010″
Soundboard thickness	02, 03	0.375″	± 0.005″
Rabbet depth	02, 06	0.250″	± 0.005″
Rabbet width	02, 06	0.375″	± 0.005″
Tongue length (ding A3)	03	6.454″	+ 0.030″ initial
Tongue width	03	1.250″	± 0.020″
Slit kerf width	03	0.125″	± 0.005″
Gu port diameter (final)	02, 09	2.500″ nominal	tuned in shop
Stave miter angle	04	11.25°	± 0.1°

Sheet 03 — tongue layout details

The 11-tongue A Kurd field uses a **zigzag-ascending** pattern: ding centered, then tongues 1, 2, 3, … alternating left and right at progressively greater radii from the ding. This matches steel-tongue-drum convention and gives the player an ergonomic two-handed reach.

Position grid (radial coords, ding = origin):

#	Note	r (in from ding centerline)	θ (deg)	length (in)
Ding	A3	0	—	6.454
1	E4	4.0	60	5.273
2	F4	4.0	-60	5.123
3	G4	5.5	90	4.835
4	A4	5.5	-90	4.564
5	B4	6.5	120	4.308

| 6 | C5 | 6.5 | -120 | 4.185 |

| 7 | D5 | 6.0 | 30 | 3.950 |

| 8 | E5 | 6.0 | -30 | 3.728 |

| 9 | G5 | 6.5 | 150 | 3.419 |

| 10 | A5 | 6.5 | -150 | 3.227 |

(These radial coords are derived estimates for the layout drawing; SolidWorks parametric placement against shell radius will refine them. The radii are kept ≤ 6.5″ to leave wall clearance ≥ 1.0″ on the 16″ shell.)

Status

These SVG files are listed in this brief as the *intended* drawing set. The repo currently ships the brief and the source workbook; SVGs are emitted by `scripts/generate_drawings.py` (in the parent `instrument-maker` skill) reading the workbook as its parametric source. Per the v4 deliverable list, drawings are auto-generated and refreshed when the workbook changes — they are not hand-drawn artifacts.

A placeholder hero drawing — `drawings/00-hero-v1-standard.svg` — ships in this repo at first commit so cross-references resolve. It is replaced in production by sheet 02 above.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Wood Shell Tongue Drum — Assembly Manual

Covers all four manufacturing variants (V1, V2, V3, V4) across three sizes (Travel 12″, Standard 16″, Floor Pouf 20″). Steps shared across variants are written once with branching notes where they diverge. Read `design.md` first — this manual assumes you have the parametric workbook open for size-specific dimensions and the recommended first prototype is **V1 Cylinder + Flat Top, Standard 16″, A Kurd**.

Phase 0 — Shop preparation

1. Acclimate stock 2 weeks in the shop (60–70 °F, 40–55 % RH). Padauk and Black Walnut both move with seasonal humidity; soundboard cup or warp will scrap a tongue field that was previously in tune.

2. Joint and plane stock to nominal dimensions ± 0.005″.

3. Verify your saw kerf width with a test cut. The cut-list assumes a known kerf — measure yours before sizing rough stock.

4. Check the lathe — main spindle runout < 0.003″ TIR, tailstock alignment within 0.005″. A round shell amplifies any mounting wobble.

5. Verify CNC router calibration: a stock-thickness probe and tool-length probe both within ± 0.002″. Tongue slits depend on consistent depth.

Phase 1A — Cylinder shell (V1, V2)

The cylinder is the *recommended* body for the first prototype. Two construction paths share Phases 1A.4 onward.

1A.1 Stave path (preferred for first build)

1. Rough-cut 16 staves to outline + 1/8″ on each side. Stock dimensions per `cut-list.csv` (Standard 16″: 9.5″ × 3.5″ × 0.875″).

2. Bevel both edges to 11.25° using a tilting-arbor jig referenced off the stave centerline (NOT off either edge — the stave is symmetric only when freshly ripped). Test on a sacrificial board first; verify the assembled circle closes by dry-clamping 4 staves and checking the included angle.

3. Sand the inside face to 220 grit. The inside is permanent and not turnable after glue-up.

1A.2 Segmented path (alternative for polychrome aesthetic)

1. Rough-cut 96 segments (8 rings × 12 segments) at 15° miter on a table-saw miter sled.

2. Stack rings on a lazy-susan jig for sequential glue-up. Each ring closes independently before stacking.

3. Diamond / 5-10-5 species patterns (Walnut + Maple + Cherry) read beautifully on 16″ Ottoman size — see `drawing-brief.md` §08 for layout templates.

1A.3 Stave glue-up (path A)

1. Lay all 16 staves flat on a tarp, edges touching, in their final order. Tape across all joints with strong masking tape.
2. Roll the assembly into a tube. Apply a thin even bead of Titebond III to every edge.
3. Re-roll, close to a full cylinder.
4. Wrap with three nylon ratchet straps OR run two large hose clamps. Do not over-tighten — squeeze-out should bead, not gush.
5. Stand the tube on a flat plate (¾″ MDF works) and check for end-square with a framing square. Adjust strap tension to pull any out-of-round into round.
6. Cure 24 hours.

1A.4 Lathe truing (both paths converge)

1. Mount a faceplate to a 3″ waste block. Glue the waste block to a ¼″ ply disk slightly larger than the shell ID.
2. Slide the shell over the disk; insert a soft-rubber expansion plug at the open end with the tailstock for support.
3. Turn the OD true to the design diameter (e.g., 16.00″ for Standard). Wall thickness target: 0.625″ finished.
4. Reverse-mount and turn the rim end. The rim must be planar (± 0.005″ TIR) — this is the airtight bond surface for the soundboard.
5. Cut a ¼″-deep × ■″-wide rabbet in the inside of the rim, sized to receive the soundboard disc (see `cut-list.csv` row S-DISC-01 for the disc dimensions).

Phase 1B — Hemisphere bowl (V3, V4)

Pick one of two construction paths. Segmented is recommended for the first hemisphere build.

1B.1 Segmented bowl path

1. Rough-cut graduated rings per `cut-list.csv` rows S-BOWL-RING-01 through S-BOWL-RING-08 (Standard 16″: 8 rings of decreasing diameter from 16″ at the rim to ~3″ at the apex).
2. Glue each ring closed on the lazy-susan jig before stacking.
3. Stack rings with a centering jig (a vertical dowel through ring centers) and glue with Titebond III. Cure 24 h between layers.
4. Mount the assembled bowl on a faceplate with a thick waste block.
5. Lathe-true the outside to a parabolic profile (within 1/16″ of true sphere is acceptable — the lathe finishes any micro-step). Lathe-true the rim flat (± 0.005″ TIR).

1B.2 CNC-carved bowl path (alternative; high material waste)

1. Glue up a thick blank: 5 plies of 1″ Black Walnut for Standard 16″.

2. CNC 3D-surface the outside convex profile with a ¾″ ball-end bit; 0.05″ stepover for ≤ 32 µin surface finish.

3. Flip-jig the blank with datum pins; 3D-surface the inside concave profile.

4. Trim rim flat on a router table or lathe; verify ± 0.005″ TIR.

Phase 2 — Soundboard (variant-dependent)

2A — Flat soundboard (V1, V3)

1. Cut Padauk disc to OD per `cut-list.csv` (Standard 16″: 16.25″ rough → 16.00″ final).

2. Mark the tongue layout on the *inside* face of the disc. Use the layout template in `cnc/README.md`.

3. CNC-route the tongue slits with a 1/8″ upcut spiral. Cut from inside face. Tongue lengths from `design.md` §4 — cut LONG by 0.030″ for tuning margin.

4. Sand the soundboard to 320 grit. **Do not break the slit edges** — chip damage causes pitch shifts that are difficult to recover.

5. Apply Watco Danish Oil to the *outside* face only. Three coats, 24 h between. Inside cavity stays unfinished (sealing the cavity damps the bass).

2B — Domed soundboard (V2, V4)

1. Glue up a 3-ply blank per `cut-list.csv` row S-DOME-BLANK (Standard 16″: 3 × 0.5″ Padauk = 1.5″ thick × 17″ × 17″).

2. **Rehearse on poplar first.** Cut a poplar test blank with the same dome profile (S-DOME-CNC-PROTO). Verify the toolpath produces a clean parabolic surface with no scallop steps before committing the Padauk blank.

3. 3D-surface the convex top with a ¾″ ball-end bit, 0.05″ stepover. Dome rise per size preset (Standard: 0.75″).

4. Flip the blank into a flip-jig with datum pins. 3D-surface the concave underside (matches OD rim for clean rabbet seat).

5. **Validate the curved cantilever multiplier.** Cut ONE tongue at the predicted length × 1.025 multiplier. Tap-test, measure pitch with a Korg OT-120. Back-solve the actual multiplier (`L_measured² × f / (K × t)` returns the new multiplier). Apply this measured multiplier to the remaining 10 tongue layouts before any further cutting.

6. Cut the remaining 10 tongues with the validated multiplier. Cut LONG by 0.030″ for tuning margin.

7. Sand and finish per Phase 2A steps 4–5.

Phase 3 — Bowl-top assembly

The bowl-to-soundboard joint is **acoustically critical**: an air leak here destroys the Helmholtz Q. Any of three joint geometries can work; the rabbet is the working assumption.

3.1 Rabbet joint (preferred)

1. Verify the shell rim rabbet (cut in Phase 1A.4 / 1B): ¼″ deep × ■″ wide, ± 0.005″.

2. Verify the soundboard slip-fits into the rabbet with 0.005″ clearance — tight enough for a PVA bond, loose enough that you don't pry the rim apart on insertion.

3. Apply a uniform thin bead of Titebond III in the rabbet. Drop the soundboard in flush.

4. Apply two cauls (top and bottom) with band clamps OR a vacuum bag. Cure 24 h.

3.2 Lap step joint (backup)

If the rabbet leaks under puff-test (Phase 6), step-lap a thin (1/16″) Padauk veneer overlapping the rabbet seam from inside. Glue with Titebond III.

3.3 Cork gasket (last-resort backup)

If both rabbet and lap-step leak, install a 1/16″ cork ring in the rabbet under the soundboard. Compresses for an airtight seal but adds compliance that may damp the Helmholtz Q.

Phase 4 — Gu-port tuning

The gu port is your primary Helmholtz tuning knob. It is drilled **last** and **deliberately undersized**, then opened up incrementally while measuring `f_H`. Opening the port raises `f_H`; adding a restrictor/liner or lengthening the port neck lowers `f_H`.

1. Drill a ¼″ pilot hole through the shell/bowl bottom, centered (verify ± 0.025″).

2. Place a small microphone above the port. Puff gently across the port edge (like blowing across a bottle top). Record the resonance frequency in REW or Spectroid.

3. Use a step-drill to enlarge the port in ¼″ increments. Re-measure `f_H` at each step. Stop when `f_H ≈ 0.88 × f_ding` (Standard 16″ V1: target ≈ 195 Hz).

4. **If the cavity is high** (ratio > 1.2 before the port reaches the target diameter): stop drilling. Lower `f_H` with a removable restrictor, port sleeve, longer effective neck, or revised chamber volume on the next build.

5. **If the cavity is low** (ratio < 0.8 even at the maximum safe port diameter): the chamber/neck model is under-predicting the port area required. Document the measured curve, do not keep enlarging into the rim margin, and re-evaluate the V2/V4 cavity model.

Optional: install a Buna-N O-ring in the finished port (BOM rows ORING-RUBBER-*) for a Hapi-style cleaner tone. Not required.

Phase 5 — Tongue tuning

Tongues are cut LONG. Each one is tuned by filing the tongue tip (which shortens the effective length, *raising* the pitch). You cannot lower a tongue's pitch by filing — if a tongue measures sharp of target by more than 50 cents at first cut, that tongue is scrap.

1. Set up the drum on a soft surface (foam, blanket).

2. Strike each tongue at its tip with a soft mallet. Record pitch with a Korg OT-120 (or a phone tuner with cents resolution).

3. Compare to `validation.csv` predicted column.

4. For tongues > 5 cents flat: do nothing yet (let the tongue settle for 48 h after cutting before tuning).

5. For tongues > 5 cents sharp: file the tongue tip with a flat sanding bar (320 grit). Test pitch every 3–4 strokes. Each 0.030″ removed from the tip raises pitch by ~10–15 cents.

6. For tongues > 30 cents flat after 48 h settling: file the *underside* of the tongue at the base (thinning the cantilever lowers stiffness, which lowers pitch). This is a recovery move; ideally tongues are cut long enough to never need it.

7. Final tuning: ± 5 cents of target on all 11 tongues.

8. Document final measurements in `validation.csv`.

Phase 6 — Validation and acceptance

Run the full validation loop before declaring the build complete:

Check	Tool	Acceptance band
All 11 tongues to target	Korg OT-120	± 5 cents per tongue
Helmholtz coupling ratio	mic + spectrum	$0.80 \le f_H/f_{ding} \le 1.20$
Bowl-top airtight seal	smoke or vacuum test	no visible leakage at rabbet
Bearing-edge planarity	dial indicator	± 0.005″ TIR
Wall thickness uniformity	calipers, 8 points	± 0.030″ from nominal
Soundboard thickness uniformity	calipers, 9-pt grid	± 0.005″
Slit kerf width	feeler gauge, 5-pt	0.125″ ± 0.005″
Sustain T60	REW software	1.5 s ≤ T60 ≤ 4.0 s
A4 = 440.0 Hz sanity	tongue 4 measurement	within 5 cents of 440 Hz

Record measured values in `validation.csv`. If A4 measures off > 5 cents, back-compute the actual K-constant of your specific stock (`K_actual = f × L² / t`) and update the per-family corrections database.

Phase 7 — Finish and presentation

1. Apply 3 coats Watco Danish Oil to the shell exterior (24 h between coats).

2. Apply Renaissance Wax to the shell exterior (final coat, buffed).

3. Apply 3 coats Watco Danish Oil to the soundboard *outside* face only.

4. Adhere felt foot pads to the shell bottom (4 pads at 90° spacing).
5. Optional: install hemp carry handle (Standard size and above).
6. Photograph per `photo-shotlist.md`.
7. Tag the drum with maker plate (interior) and update the catalog entry in the master workbook.

Branching summary

Phase	V1 (Cyl + Flat)	V2 (Cyl + Dome)	V3 (Hemi + Flat)	V4 (Hemi + Dome)
1	1A — cylinder	1A — cylinder	1B — bowl	1B — bowl
2	2A — flat	2B — dome	2A — flat	2B — dome
3	3.1 rabbet	3.1 rabbet	3.1 rabbet	3.1 rabbet
4	2.5″ port	larger measured port or sleeve experiment	2.5″ port	larger measured port or sleeve experiment
5	flat tuning	flat + dome correction	flat tuning	flat + dome correction
6	full validation	full validation + dome multiplier check	full validation	full validation + dome multiplier check
7	standard finish	standard finish	standard finish	standard finish

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

size	variant	target	parameter	predicted_value	measured_value	unit	cents_error	measurement_date	instrument	notes
Standard 16in	V1	ding fundamental	frequency	220.00		Hz			Korg OT-120	A3 ding (MIDI 57). Tunable b
Standard 16in	V1	tongue 1 (E4)	frequency	329.63		Hz			Korg OT-120	E4 tongue. Cut 0.030 in long
Standard 16in	V1	tongue 2 (F4)	frequency	349.23		Hz			Korg OT-120	
Standard 16in	V1	tongue 3 (G4)	frequency	392.00		Hz			Korg OT-120	
Standard 16in	V1	tongue 4 (A4)	frequency	440.00		Hz			Korg OT-120	Reference pitch — should lar
Standard 16in	V1	tongue 5 (B4)	frequency	493.88		Hz			Korg OT-120	
Standard 16in	V1	tongue 6 (C5)	frequency	523.25		Hz			Korg OT-120	
Standard 16in	V1	tongue 7 (D5)	frequency	587.33		Hz			Korg OT-120	
Standard 16in	V1	tongue 8 (E5)	frequency	659.26		Hz			Korg OT-120	
Standard 16in	V1	tongue 9 (G5)	frequency	783.99		Hz			Korg OT-120	
Standard 16in	V1	tongue 10 (A5)	frequency	880.00		Hz			Korg OT-120	
Standard 16in	V1	Helmholtz cavity	frequency	194.57		Hz			microphone + s	Cavity port puff test. Target 0
Standard 16in	V1	coupling ratio	f_H/f_ding	0.88		ratio			computed	Acceptance band 0.80-1.20 f
Standard 16in	V1	sustain T60	RT decay -60dB	2.5		seconds			REW software	Acceptance 1.5-4.0 s. Wood
Standard 16in	V1	bearing-edge planarity	TIR	0.005		inches			dial indicator	Lathe-tested. Acceptance +/-
Standard 16in	V1	wall thickness uniformity	8 points around shell	0.625		inches			calipers	+/- 0.030 in acceptance. Wed
Standard 16in	V1	soundboard thickness uniformity	9-pt grid	0.375		inches			calipers	+/- 0.005 in acceptance. Affe
Standard 16in	V1	slit kerf width	5-point sample	0.125		inches			feeler gauge	+/- 0.005 in. Wider kerfs = lov
Standard 16in	V1	gu-port concentricity	distance from bowl center	0		inches			calipers	+/- 0.025 in. Off-center port s
Standard 16in	V1	bowl-top airtight seal	pressurization test	pass		pass/fail			vacuum test or smoke	No air leakage at the rabbet j
Standard 16in	V1	glue-line shear (24hr cure)	destructive sample test	4000		psi			test fixture	Titebond III rated 4000 psi. T
Travel 12in	V1	ding fundamental	frequency	293.66		Hz			Korg OT-120	D4 ding (MIDI 62). Build last
Travel 12in	V1	Helmholtz cavity	frequency	321.53		Hz			microphone + s	Predicted ratio 1.09 — at upp
Travel 12in	V1	coupling ratio	f_H/f_ding	1.09		ratio			computed	Tight; acceptance 0.80-1.20.
Floor Pouf 20in	V1	ding fundamental	frequency	146.83		Hz			Korg OT-120	D3 ding. WARNING: ding ton
Floor Pouf 20in	V1	Helmholtz cavity	frequency	144.69		Hz			microphone + s	Predicted ratio 0.99 — perfec
Floor Pouf 20in	V1	coupling ratio	f_H/f_ding	0.99		ratio			computed	Sweet spot.
Standard 16in	V2	ding fundamental	frequency	220.00		Hz			Korg OT-120	A3 ding. Domed top requires
Standard 16in	V2	curved-cantilever correction	measured/predicted ratio	1.025		multiplier			Korg OT-120	VALIDATE EMPIRICALLY: ta
Standard 16in	V2	Helmholtz cavity	frequency	133.48		Hz			microphone + s	Predicted ratio 0.61 — UNDE
Standard 16in	V3	ding fundamental	frequency	220.00		Hz			Korg OT-120	A3 ding, hemisphere bowl, fla
Standard 16in	V3	Helmholtz cavity	frequency	238.30		Hz			microphone + s	Predicted ratio 1.08 — couple
Standard 16in	V3	coupling ratio	f_H/f_ding	1.08		ratio			computed	Coupled.
Standard 16in	V4	ding fundamental	frequency	220.00		Hz			Korg OT-120	
Standard 16in	V4	Helmholtz cavity	frequency	161.12		Hz			microphone + s	Predicted ratio 0.73 — UNDE
All	All	A4 = 440 Hz sanity	reference pitch	440.00		Hz			Korg OT-120	Sanity check: tongue 4 (A4) i
All	All	Padauk K calibration	measured K constant	24438		K imperial			back-computed from tuning	After tongue measurement, back-compute K
All	V2+V4	curved cantilever K-effective	measured/predicted ratio	1.025		multiplier			first dome prototype	Per-family corrections databa

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Wood Shell Tongue Drum (Q2 2026 batch)

Heifer Zephyr Instruments is producing a wood shell tongue drum series across four manufacturing variants (V1–V4) and three sizes (Travel 12″, Standard 16″, Floor Pouf 20″). The recommended first prototype is **V1 Cylinder + Flat Top, Standard 16″** — that build is the priority for this RFQ. Variants V2–V4 are queued for sequential phases and should be quoted alongside but not necessarily shipped together.

Verified pricing date for all line items: **2026-05-07**. All prices are starting-point estimates; supplier confirmation requested before commit.

Lumber

Need	Quantity (per drum)	Specification
Black Walnut, 4/4, S2S, FAS	~3 BdFt (Standard 16″)	Edge-jointable for stave glue-up; clear grain
Black Walnut, 4/4, S2S, FAS	~5 BdFt (segmented bowl)	Bulk segment grade; 12-segment ring stack
Padauk, 4/4 or 1/2, S4S	~1 BdFt	Soundboard thickness controls tongue length — hold 3/8″ ± 1/64″ on Standard. Quartersawn preferred for bend stability.
Padauk, 4/4 boards (3 ea, glue-up)	~3 BdFt (V2/V4 dome only)	For glue-laminated dome blank; 1.5″ total thickness
Western Red Cedar, 1/2 quartersawn	~1 BdFt (Floor Pouf alt)	Soundboard alternate for D3 ding fit on Floor Pouf 20″
Poplar, 1× boards	~3 BdFt (V2/V4 prototype)	Cheap test stock for dome surfacing rehearsal

Preferred suppliers: Bell Forest Products (primary), Cook Woods (segment blanks). Local pickup acceptable if equivalent grade.

Tooling

For shop and CNC tooling. Most are one-time buys; the 1/8″ upcut spirals are consumables (one breaks per shell).

Item	Qty	Source
Solid carbide upcut spiral 1/8″ × 1/4″ shank	3	Amana 46202 (preferred)
Ball-end mill 3/4″ × 1/2″ shank (V2/V4 dome surfacing)	1	Amana 46377
Rabbeting bit set with bearing kit, 1/4″ shank	1	Rockler
Step drill bit, 1/4″–3″ in 1/4″ increments	1	Irwin Unibit (Amazon)

Hardware and consumables

Item	Qty (per drum)	Notes
Titebond III wood glue, 16 oz	1 (shared 2 drums)	Waterproof; 4000 psi shear
Watco Danish Oil, natural, qt	1 (shared 4 drums)	3-coat finish
Renaissance Wax, 65 ml	1 (shared 4 drums)	Final shell finish
Sandpaper assortment 80/120/220/320/400/600 grit	1 set	Local hardware store
Buna-N O-ring (sized to gu port)	1	Optional Hapi-style port lining
Adhesive felt floor-protection pads, 1″ dia	4	Local hardware
Hemp rope, 1/2″, 4 ft	1 (Std/Floor)	Optional carry handle

Skin/heads

N/A — this is a slit-tongue idiophone, not a membrane drum. No skins required. (Distinguishes this design from the conga and ashiko families which use mule or LP heads.)

Acoustic test gear (one-time shop equipment)

These are workshop instruments, not per-drum consumables. Listed for budget transparency.

Item	Qty	Notes
Korg OT-120 wide-range chromatic tuner	1	Already owned
Microphone + REW software for Helmholtz	1	Phone + free Spectroid app acceptable

Lead-time summary

Category	Typical lead time
Lumber (Bell Forest)	5 days
Lumber (Cook Woods)	4 days
Tooling (Amana, Rockler)	2–3 days
Hardware (McMaster)	2 days
Local hardware	same-day

Critical path for first prototype: lumber + slit bits. Order both at start of Phase 1; tooling can be ordered immediately on project kickoff.

Quote request

Please confirm:

1. Current per-board-foot price on Black Walnut 4/4 FAS (we want clear, edge-jointable stock — figured grade is *not* required for this design and adds cost without acoustic benefit).
2. Padauk availability in 3/8″ thickness *as a stock item*. If only 4/4 is stocked, we will glue / resaw — quote both options.
3. Lead-time guarantees for Q2 2026.
4. Volume-discount thresholds — we may run multiple drums (V1 first, V2 reuses the same shell) over the next 12 months.

Heifer Zephyr Instruments

Tony Koop

github.com/tonykoop/wood-shell-tongue-drum

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Visual BOM — Photo shotlist for the BOM hero plate

A Visual BOM lays out every part for one drum on a flat surface against a clean backdrop and labels the parts with quantities, IDs, and final cost. The result is a single hero-style photograph that serves as the cover image for the build-log site, the printable shop packet, and the recruiter-facing deck.

Hero shot — V1 Standard 16″ (recommended first prototype)

Frame: top-down, 4:5 portrait orientation, shop-tabletop or seamless butcher-paper backdrop (warm neutral — kraft or off-white).

Lighting: one diffused softbox at 45° camera-left, slight bounce-fill from camera-right.

Layout (top-down grid, parts arranged left-to-right by build phase):

Row 1 — Body

[16 stave blanks fanned in a circle] [shell after lathe-turning, set on rim] [rabbet detail close-up]

Row 2 — Soundboard

[Padauk disc with slit pattern routed] [1/8" upcut spiral bit] [tongue length schedule card]

Row 3 — Tuning hardware (sparse)

[step drill bit] [Buna-N O-ring] [Korg OT-120 tuner]

Row 4 — Finish + accessory

[Watco Danish Oil] [Renaissance Wax] [pair of mallets] [felt foot pads]

Caption: `Wood Shell Tongue Drum — V1 Cylinder + Flat Top, Standard 16″ — Heifer Zephyr Instruments — Total BOM ~$XXX`. Plug the actual rolled-up sum from `bom.csv` extended_cost column for V1+Standard rows.

Detail shots — shared across variants

1. **Stave miter** — close-up of one stave edge showing 11.25° bevel against a digital protractor. Side-lit to emphasize the angle.

2. **Tongue slit** — extreme close-up of one slit-and-tongue, top-down, showing kerf width and the slit terminus geometry. Side-lit.

3. **Rabbet joint** — section view of the rabbet with the soundboard half-engaged. Highlights the air-seal surfaces.

4. **Gu port** — bottom view of the shell with the step-drilled port; ruler in frame for scale.

5. **Bowl ring stack** (V3/V4 only) — one of the graduated rings photographed flat against the backdrop, showing 12-segment construction with the 15° miters.

6. **Domed soundboard** (V2/V4 only) — 3/4 angle showing the dome rise on the Padauk blank after CNC surfacing; light raking across the curve to show the parabolic profile.

Variant comparison shot (after Phases 1–4)

After all four variants are built, one final group shot:

• V1, V2, V3, V4 in a row at Standard 16″ size, soundboards facing camera, 3:2 landscape orientation.

• Backdrop: dark-gray seamless or wood floor.

• Lighting: two softboxes at 45° each side, slightly above eye level.

• Caption: `Heifer Zephyr Wood Shell Tongue Drum — V1 → V4 Family — All Standard 16″, A Kurd`.

This becomes the hero on the build-log site (`site/index.html`) once all four variants ship.

Process / progression montage

A 4 × 2 grid of in-shop process shots illustrating the build sequence:

1. Stave glue-up under ratchet straps
2. Lathe-truing the OD
3. CNC routing the tongue slits in Padauk soundboard
4. Rabbet joint assembly with cauls
5. Gu-port step-drilling with mic + tuner in frame
6. Tongue tuning with sanding bar and Korg
7. Final finish — Danish Oil application
8. Hero shot of the finished V1 Standard

This montage anchors the build-log site mid-page and serves as the printable packet's "build workflow" foldout.

Constraints

• All photographs must show **the actual instrument and parts**, not concept renders. The concept sheet in `concepts/` is for ideation; the BOM hero is for documentation.

• Dimensions in caption text must match `bom.csv` and `cut-list.csv` exactly. Any drift is a packet-validation failure.

• No props that aren't part of the BOM. Parts should look like a clean inventory, not a styled lifestyle shot.

• Shot list updates: as builds complete, replace each Required cell with a real path (`images/IMG-XXXX.jpg`) and mark Status = Captured.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

```
(* ::Package:: *)

(* Wood Shell Tongue Drum — Acoustic Physics Starter
Heifer Zephyr Instruments — Tony Koop
Open in Wolfram Desktop or Wolfram Cloud.

This notebook implements the 3-DOF coupled-oscillator picture for a
slit-tongue idiophone with an enclosed Helmholtz cavity. The three
oscillators are:

1. The tongue (flat or curved cantilever, Euler-Bernoulli).
2. The shell wall (cylindrical or hemispherical).
3. The enclosed air cavity (Helmholtz resonator).

The tongue couples to the shell through the rabbet boundary; the
shell couples to the air cavity through wall compliance; the air
cavity couples back to the tongue through pressure on the
underside of the slit field. The eigenvalue split predicts how much
the wood shell adds sustain compared to a rigid-walled tongue drum.
*)

(* === 1. Inputs (edit me) === *)

(* All dimensions in inches; conversions to SI happen below. *)

(* Recommended first prototype: V1 Standard 16in *)
shellOD = 16.0; (* shell outer diameter *)
shellHeight = 8.0; (* cylinder body height *)
shellThk = 0.625; (* wall thickness, finished *)
sbThk = 0.375; (* soundboard thickness (Padauk) *)
sbWood = "Padauk"; (* one of Padauk, Cherry, Walnut, Maple, Cedar *)
shellWood = "BlackWalnut"; (* shell material *)
guPortDia = 2.5; (* gu port diameter, initial preset *)
domeRise = 0.0; (* 0 for flat top, 0.75 for V2 Std *)

dingMidi = 57; (* A3 *)
nTongues = 11;
```

```
tongueWidth = 1.25;
slitKerf = 0.125;

(* Material library — cantilever K (imperial composite) and elastic
moduli. Cross-references the workbook material library, rows 19-25. *)
materials = <|
"Padauk" -> <|"K" -> 24438., "E" -> 11.0*^9, "rho" -> 745.|>,
"Wenge" -> <|"K" -> 27103., "E" -> 15.8*^9, "rho" -> 870.|>,
"Cherry" -> <|"K" -> 27275., "E" -> 10.3*^9, "rho" -> 560.|>,
"BlackWalnut" -> <|"K" -> 27734., "E" -> 11.6*^9, "rho" -> 610.|>,
"HardMaple" -> <|"K" -> 26887., "E" -> 12.6*^9, "rho" -> 705.|>,
"Mahogany" -> <|"K" -> 26314., "E" -> 10.1*^9, "rho" -> 590.|>,
"WesternRedCedar" -> <|"K" -> 29013., "E" -> 7.7*^9, "rho" -> 370.|>
|>;

(* === 2. Constants === *)

cAir = 13552.; (* speed of sound in inches/sec *)
mPerIn = 0.0254; (* unit conversion *)

(* MIDI -> Hz *)
fFromMidi[m_] := 440. * 2.^((m - 69)/12.);

(* === 3. Tongue cantilever (DOF 1) === *)

(* Flat cantilever Euler-Bernoulli: f = K * t / L^2 => L = sqrt(K * t / f) *)

LFlat[K_, t_, f_] := Sqrt[K * t / f];
LCurved[K_, t_, f_, mult_:1.025] := mult * Sqrt[K * t / f];

(* Predicted ding tongue length, V1 vs V2 *)
KSb = materials[sbWood, "K"];
fDing = fFromMidi[dingMidi];

LDingFlat = LFlat[KSb, sbThk, fDing];
LDingCurved = LCurved[KSb, sbThk, fDing];

(* Print *)
Print["Predicted tongue lengths for ", sbWood,
" soundboard, ", sbThk, " in thick:"];
Print[" Ding (MIDI ", dingMidi, ", ", NumberForm[fDing, 5], " Hz):"];
```

```
Print[" flat = ", NumberForm[LDingFlat, 4], " in"];
Print[" curved = ", NumberForm[LDingCurved, 4], " in (multiplier 1.025)"];

(* === 4. Helmholtz cavity (DOF 3) === *)

(* fH = (c / 2 pi) * sqrt(A / (V * Lneck)) *)

fHelmholtz[V_, A_, Lneck_] := (cAir / (2 Pi)) * Sqrt[A / (V * Lneck)];

(* Volume formulas per variant (in cubic inches) *)
volV1 = Pi * (shellOD/2)^2 * shellHeight; (* cyl flat *)
volV2 = volV1 + (2/3)*Pi*(shellOD/2)^2 * domeRise; (* cyl + dome *)
volV3 = (2/3) * Pi * (shellOD/2)^3; (* hemi flat *)
volV4 = volV3 + (2/3)*Pi*(shellOD/2)^2 * domeRise; (* hemi + dome *)

portArea = Pi * (guPortDia/2)^2;
neckFlat = sbThk;
neckDomed = sbThk + domeRise/2;

fHV1 = fHelmholtz[volV1, portArea, neckFlat];
fHV2 = fHelmholtz[volV2, portArea, neckDomed];
fHV3 = fHelmholtz[volV3, portArea, neckFlat];
fHV4 = fHelmholtz[volV4, portArea, neckDomed];

Print[];
Print["Helmholtz frequencies (current preset, gu port = ", guPortDia, " in):"];
Print[" V1 (cyl + flat) : ", NumberForm[fHV1, 5], " Hz, ratio ",
NumberForm[fHV1/fDing, 3]];
Print[" V2 (cyl + dome) : ", NumberForm[fHV2, 5], " Hz, ratio ",
NumberForm[fHV2/fDing, 3]];
Print[" V3 (hemi + flat) : ", NumberForm[fHV3, 5], " Hz, ratio ",
NumberForm[fHV3/fDing, 3]];
Print[" V4 (hemi + dome) : ", NumberForm[fHV4, 5], " Hz, ratio ",
NumberForm[fHV4/fDing, 3]];
Print[" Coupled regime: 0.80 <= ratio <= 1.20"];

(* === 5. 3-DOF coupled-oscillator model === *)

(*
Reduced model: tongue (m1, k1), shell wall (m2, k2), air (m3, k3)
coupled by k12 (rabbet stiffness) and k23 (wall-air compliance).
```

```
Solving the 3x3 mass-spring system yields three eigenfrequencies;
the lower two are dominated by tongue and Helmholtz, the highest
by the shell wall mode. The split between the tongue eigenfreq
and the lone-tongue freq quantifies the shell's contribution to
sustain.

The values below are starting estimates -- they are NOT measured.
Treat the model as a structural starting point; the per-family
corrections database holds the calibrated numbers once measurements
exist.
*)

(* Effective masses (kg) and stiffnesses (N/m) -- order-of-magnitude *)
(* Tongue: half-beam approximation, treat as single-DOF cantilever *)
m1 = 0.020; (* ~20 g effective tongue mass *)
k1 = (2 Pi fDing)^2 * m1;

(* Shell wall: ring-mode mass, large stiffness *)
m2 = 0.5; (* ~500 g effective wall mass *)
k2 = (2 Pi 800.)^2 * m2; (* assumed ring mode ~800 Hz *)

(* Air cavity: lumped Helmholtz oscillator *)
m3 = 0.01; (* effective air-plug mass at port *)
k3 = (2 Pi fHV1)^2 * m3;

(* Coupling stiffnesses *)
k12 = 0.10 * k1; (* 10% of tongue stiffness through rabbet *)
k23 = 0.05 * k3; (* 5% wall-air through compliance *)

KMat = {
{k1 + k12, -k12, 0.0 },
{-k12, k2 + k12 + k23, -k23},
{0.0, -k23, k3 + k23}
};
MMat = DiagonalMatrix[{m1, m2, m3}];

eigSystem = Eigensystem[N[Inverse[MMat] . KMat]];
omegas = Sqrt[eigSystem[[1]]];
freqs = Sort[omegas / (2 Pi)];
```

```
Print[];
Print["3-DOF coupled-oscillator eigenfrequencies (V1 preset):"];
Print[" Lone tongue : ", NumberForm[fDing, 5], " Hz"];
Print[" Coupled f1 : ", NumberForm[freqs[[1]], 5], " Hz (Helmholtz-dominated)"];
Print[" Coupled f2 : ", NumberForm[freqs[[2]], 5], " Hz (tongue-dominated)"];
Print[" Coupled f3 : ", NumberForm[freqs[[3]], 5], " Hz (wall-mode-dominated)"];
Print["Sustain hint: closer f1 and f2 -> stronger bass/mid coupling."];

(* === 6. Manipulate: tune the cavity in real time === *)

(*
Open this in Wolfram Desktop. Drag gu port size; watch the
Helmholtz / ding ratio cross the coupled band.
*)

manipulateCell = Manipulate[
Module[{A, V, Lneck, fH, ratio},
A = Pi (gu/2)^2;
V = If[variant == "V1", Pi (shellOD/2)^2 hh,
If[variant == "V2", Pi (shellOD/2)^2 hh + (2/3) Pi (shellOD/2)^2 dr,
If[variant == "V3", (2/3) Pi (shellOD/2)^3,
(2/3) Pi (shellOD/2)^3 + (2/3) Pi (shellOD/2)^2 dr]]];
Lneck = If[variant == "V1" || variant == "V3", sbThk, sbThk + dr/2];
fH = (cAir/(2 Pi)) Sqrt[A/(V Lneck)];
ratio = fH / fFromMidi[ding];
Column[{
Row[{"Helmholtz f_H = ", NumberForm[fH, 5], " Hz"}],
Row[{"Ding = ", NumberForm[fFromMidi[ding], 5], " Hz"}],
Row[{"Ratio = ", NumberForm[ratio, 3],
If[0.80 <= ratio <= 1.20, " <- coupled", " <- mistuned"]}]
}]
],
{{variant, "V1"}, {"V1", "V2", "V3", "V4"}, ControlType -> RadioButtonBar},
{{ding, 57, "ding MIDI"}, 50, 70, 1},
{{gu, 2.5, "gu port (in)"}, 0.5, 4.0, 0.05},
{{hh, 8.0, "shell height (in)"}, 4.0, 12.0, 0.1},
{{dr, 0.75, "dome rise (in)"}, 0.0, 1.5, 0.05}
```

```
];

(* manipulateCell *) (* uncomment to display *)

(* === 7. Audio synthesis (preview) === *)

(* AudioGenerator can render the predicted tongue field; useful for
pre-build pitch-target previews. *)

scaleAKurd = {220.00, 329.63, 349.23, 392.00, 440.00, 493.88, 523.25,
587.33, 659.26, 783.99, 880.00};

(* tonguePulse[freq] returns a short attack-decay envelope over a sine
approximation of the tongue's first mode *)
tonguePulse[freq_, dur_:0.6] :=
Sound[SoundNote["", dur, "FluteSection", SoundVolume -> 0.6, freq]];

(* Uncomment to play the predicted A Kurd scale: *)
(* AudioPlay[Sound[Table[tonguePulse[f], {f, scaleAKurd}]]] *)

(* === 8. Notes for the per-family corrections database === *)

(*
When the first prototype is tuned, run scripts/record_measurement.py
from the parent instrument-maker skill. Each measurement updates:

- K_constant for sbWood (per stock batch)
- curved-cantilever multiplier (V2/V4 only, per dome rise)
- end-correction term for Helmholtz neck length (per shell type)

These propagate to every sibling packet in the wood-shell-tongue-drum
family on next packet generation.
*)
```

README.md

Project artifact.

Wood Shell Tongue Drum

Slit-tongue idiophone with an enclosed wooden resonator. A round-body counterpart to the rectangular-prism wood tongue drum, with **Helmholtz cavity coupling** absent from open designs. Four geometric variants × three size envelopes, all parametric and traceable to a single design table.

Part of the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue.

`![V1 Standard 16″ side section A-A](drawings/00-hero-v1-standard.svg)`

V1 Cylinder + Flat Top, Standard 16″ — recommended first prototype. Black Walnut staves, Padauk soundboard, A Kurd 11-tongue field, gu port for Helmholtz tuning. Computed `f_H = 194.6 Hz` sits at 0.88 × the A3 ding (220 Hz) — inside the coupled regime by design.

What's new in this design

A round-body wood tongue drum with an enclosed cavity is not a commercial category at this depth. Three things distinguish it:

1. **Round body, not rectangular.** Tony's existing rectangular-prism wood tongue drum is the precursor; this design moves the same flat-cantilever physics onto a turned cylinder or hemisphere bowl.

2. **Enclosed Helmholtz cavity.** Open rectangular tongue drums waste cavity volume out the open ends. Closing the bowl puts a Helmholtz resonator under the tongue field — bass extension below the lowest tongue, tunable in the shop via the gu port.

3. **Premium tonewoods, manufacturable scope.** Padauk soundboard for the A4 reference; Black Walnut shell for the bowl. Four variants × three sizes laid out as a coordinated portfolio family rather than four bespoke pieces.

The matrix

Four geometric variants × three size envelopes = twelve cells. Each cell carries its own chamber volume, Helmholtz frequency, tongue-fit verdict, and risk grade.

Variant	Body	Top	Risk	Notes
V1	Cylinder (ottoman silhouette)	Flat soundboard	★★	Lowest piece-count, most predictable tuning. **Start here.**
V2	Cylinder	Shallow CNC-domed	★★★	Steel-tongue-drum aesthetic on a wooden body
V3	Hemisphere bowl	Flat soundboard	★★	Tagine / Moroccan-pouf silhouette
V4	Hemisphere bowl	Domed top	★★★★	Most steel-tongue-drum-adjacent silhouette in wood

Sizes: Travel 12″ Ø · Standard 16″ Ø (Moroccan-ottoman scale) · Floor Pouf 20″ Ø.

The full Helmholtz/ding ratio matrix is in [`design.md`](design.md) §2.4. Three cells sit cleanly in the coupled regime (V1 Standard, V1 Floor Pouf, V3 Standard); V2 cells need a larger-than-preset gu port or a lower cavity/neck correction to reach coupling; V1 Travel sits right at the upper edge of the band.

Recommended first prototype

V1 Cylinder + Flat Top, Standard 16″, A Kurd, A3 ding (220 Hz), 11 tongues. Justification:

- Acoustic predictability — flat-cantilever physics has < 1 % empirical detuning on Tony's prior rectangular drums.
- Helmholtz coupled out-of-the-box at `f_H/f_ding = 0.88`.
- All 11 A Kurd tongues fit within the 7″ radial cap with 8 % margin.
- Lowest piece-count: 16 staves + 1 disc = 17 wood pieces.
- Dual body-construction paths (stave or segmented) — pick the aesthetic, the physics is the same.
- Validated workflow components — stave glue-up, lathe truing, CNC slit routing — exercised on Tony's existing ashiko, conga, and tongue-drum repos.

Detailed prototype spec (dimensions, tongue length schedule, joint tolerances) in [`design.md`](design.md) §4.

Acoustic model in one paragraph

The wood shell tongue drum is a 2-DOF coupled oscillator: the cantilever tongue is one resonator, the enclosed air cavity is another, and they share a common boundary (the soundboard). Tongues follow flat-cantilever Euler-Bernoulli $f = K \cdot t / L^2$, with `K ≈ 24,438` for Padauk. Domed tongues add a +2.5 % length correction (a starting estimate to be calibrated empirically on the first V2 build). The cavity follows the Helmholtz formula $f_H = (c/2\pi) \cdot \sqrt{A_{port} / (V \cdot L_{neck})}$; the gu port is the primary tuning knob — drilled last and step-bored in the shop while measuring `f_H`. Target: $f_H/f_{ding} \in [0.80, 1.20]$. See [`design.md`](design.md) §2 for the full derivation.

Repo contents

File / folder	Purpose
[`design.md`](design.md)	Governing model, variant blocks, recommended prototype
[`wood-shell-tongue-drum-design-table.xlsx`](wood-shell-tongue-drum-design-table.xlsx)	Parametric workbook (190 formulas, 1 sheet)
[`bom.csv`](bom.csv) / [`sourcing.csv`](sourcing.csv) / [`cut-list.csv`](cut-list.csv) / [`validation.csv`](validation.csv)	Manufacturing CSVs
[`assembly-manual.md`](assembly-manual.md)	7-phase build instructions
[`supplier-rfq.md`](supplier-rfq.md)	Q2 2026 batch supplier RFQ
[`drawing-brief.md`](drawing-brief.md) / [`drawings/`](drawings/)	9-sheet drawing spec + hero SVG
[`visual-bom-brief.md`](visual-bom-brief.md) / [`photo-shotlist.md`](photo-shotlist.md)	Photography briefs

| [`risks.md`](risks.md) | Red-team pass with verification tests attached |

| [`wolfram-starter.wl`](wolfram-starter.wl) | 3-DOF coupled-oscillator notebook starter |

| [`capstone-deck.md`](capstone-deck.md) / [`print-packet.md`](print-packet.md) | Recruiter-facing deck + shop-floor printable packet |

| [`cad/`](cad/) / [`cnc/`](cnc/) | Toolpath plans, jig decisions, laser templates, CAD staging |

| [`cnc/jig-and-template-plan.md`](cnc/jig-and-template-plan.md) | Fixture choice matrix for the first prototype and later variants |

| [`site/index.html`](site/index.html) | Build-log static site |

| [`concepts/`](concepts/) | Original concept sheet (ideation, not manufacturable) |

Build status

• ■ Parametric design table (workbook + `design.md` packet)

• ■ BOM, sourcing, cut list, validation CSVs

• ■ Assembly manual, supplier RFQ, drawing brief, visual BOM brief, photo shot list

• ■ Risks register (16 entries across acoustic, structural, ergonomic, supply, fit/finish)

• ■ Wolfram notebook starter (3-DOF coupled oscillator)

• ■ CNC/laser/jig decision plan for public review

• ■ Build-log site (`site/index.html`)

• ■ Hero side-section drawing (`drawings/00-hero-v1-standard.svg`)

• ■ Phase 1 prototype: V1 Cylinder · Flat at 16″ Standard

• ■ SolidWorks CAD assembly (deferred until first-prototype calibration)

• ■ Auto-generated SVG sheets 01–09 from `scripts/generate_drawings.py`

• ■ Capstone markdown and print-packet PDF generated from the repo sources

• ■ Capstone .pptx refresh (requires python-pptx in the local generator environment)

Highest-risk unknowns

To be retired by Phase 1 measurements:

1. Curved-cantilever multiplier for our specific dome rise (V2/V4 only; +2.5 % is a starting estimate).
2. Helmholtz end-correction term for slit ports vs circular ports (all variants).
3. Padauk K-constant for our specific stock vs the library value (24,438).
4. Bowl-top rabbet joint long-term durability under seasonal humidity cycling.
5. Player ergonomics on Floor Pouf 20″ (new size class for Tony's catalog).

Full risk register with mitigations and verification tests in [`risks.md`](risks.md).

Sister repos

- [`tonykoop/tongue-drum`](https://github.com/tonykoop/tongue-drum) — rectangular-prism precursor, validates flat-cantilever physics
- [`tonykoop/steel-tongue-drum`](https://github.com/tonykoop/steel-tongue-drum) — steel cantilever cousin
- [`tonykoop/ceramic-tongue-drum`](https://github.com/tonykoop/ceramic-tongue-drum) — slip-cast cousin
- [`tonykoop/ashiko-drum-workshop`](https://github.com/tonykoop/ashiko-drum-workshop) — segmented bowl reference
- [`tonykoop/conga`](https://github.com/tonykoop/conga) — stave & segmented shell reference
- [`tonykoop/instrument-maker`](https://github.com/tonykoop/instrument-maker) — orchestrating skill, Master Catalog, validate_packet.py

License

photo-shotlist.md

Project artifact.

Photo Shot List — Wood Shell Tongue Drum

Generated: 2026-05-07

This shot list feeds the build-log site (`site/index.html`), the printable shop packet, and any future capstone-deck refresh. Real photographs replace SVG placeholders as builds complete. Until prototypes exist, the SVG drawings carry the visual weight of the documentation.

Required hero and overview

Shot ID	Subject	Purpose	Status
HERO-001	V1 Standard 16″ finished, 3/4 angle, on neutral backdrop	README hero, build-log site cover	Needed
HERO-002	All four variants (V1–V4) at Standard 16″ in a row, soundboards facing camera	Family lineup; print packet cover	Needed (after Phase 4)
HERO-003	Three-size ladder (Travel + Standard + Floor Pouf) of V1 only	Size-progression catalog shot	Needed (after Phase 5)
HERO-004	V1 Standard in-hand with mallets, top-down	Scale and ergonomics	Needed

Construction process — V1 Standard (recommended first prototype)

Shot ID	Subject	Purpose	Status
BUILD-101	16 stave blanks rough-cut, fanned out	Stock prep	Needed
BUILD-102	Stave on tilting-arbor jig with 11.25° bevel cut	Miter setup	Needed
BUILD-103	Staves taped flat, edges glued, ready to roll into cylinder	Stave glue-up Phase 1A.3	Needed
BUILD-104	Cylinder under three ratchet straps, squeeze-out beading	Glue-up clamp	Needed
BUILD-105	Shell mounted on lathe, OD truing pass	Phase 1A.4	Needed
BUILD-106	Rim rabbet detail, ¼″ × ■″, dial indicator showing TIR	Critical-tolerance shot	Needed
BUILD-107	Padauk soundboard disc on CNC, 1/8″ upcut spiral routing tongue slits	Phase 2A; cite slit kerf 0.125″	Needed
BUILD-108	Slit field complete on soundboard, top-down	Tongue layout reference	Needed
BUILD-109	Rabbet joint dry-fit, soundboard half-engaged	Air-seal demonstration	Needed
BUILD-110	Rabbet joint glued and clamped with cauls	Phase 3.1	Needed

BUILD-111	Gu-port step-drilling with mic + Korg in frame	Phase 4 — Helmholtz tuning	Needed
BUILD-112	Tongue tuning with sanding bar, Korg OT-120 reading	Phase 5 — final tuning	Needed

Domed-top variant (V2 Standard) — additional shots

Shot ID	Subject	Purpose	Status
DOME-201	Poplar test blank with dome surfacing complete	De-risk shot	Needed
DOME-202	Padauk dome blank in flip-jig, datum pins visible	CNC workholding	Needed
DOME-203	Domed Padauk top, raking light to show 0.75″ rise	Profile reference	Needed
DOME-204	First-tongue calibration cut, single tongue, before mass-cutting	Curved-cantilever validation	Needed
DOME-205	Korg + tap test on the calibration tongue	Multiplier measurement	Needed

Hemisphere-bowl variants (V3, V4) — additional shots

Shot ID	Subject	Purpose	Status
BOWL-301	Graduated rings stacked but unglued, showing decreasing-dia profile	Segmented bowl construction	Needed
BOWL-302	Single ring closed and glued on lazy-susan jig	Per-ring closure	Needed
BOWL-303	Stacked bowl on lathe, profile-truing pass	Lathe-truing the spherical exterior	Needed
BOWL-304	Finished bowl rim with planar TIR check	Rim flatness verification	Needed

Detail and material shots

Shot ID	Subject	Purpose	Status
DETAIL-401	Stave miter close-up against digital protractor	11.25° angle reference	Needed
DETAIL-402	Tongue slit terminus geometry, extreme close-up	Slit corner relief	Needed
DETAIL-403	Padauk grain detail on finished soundboard	Material identification	Needed
DETAIL-404	Black Walnut shell exterior with Watco Danish Oil finish, raking	Finish quality	Needed
DETAIL-405	Buna-N O-ring installed in finished gu port	Optional Hapi-style detail	Needed

Ergonomic / play context

Shot ID	Subject	Purpose	Status

PLAY-501	Player seated, V1 Standard on lap, two-mallet position	Lap-play ergonomics	Needed
PLAY-502	Player at floor with Floor Pouf, both hands striking outer tongues	Floor-play reach	Needed
PLAY-503	Side-by-side: player hands on Standard vs Floor Pouf	Size comparison for ergonomics	Needed

Audio reference

Not photo, but live audio recordings should accompany the build-log site:

Track ID	Subject	Purpose	Status
AUDIO-001	Single ding strike, decay to silence	Sustain T60 measurement	Needed
AUDIO-002	Full A Kurd scale played up	Tuning verification	Needed
AUDIO-003	A vs B vs C vs D — same scale, four variants	Variant tonal comparison	Needed (after Phase 4)
AUDIO-004	Cavity puff test recording with visible f_H peak	Helmholtz validation evidence	Needed

Status legend

• **Needed** — not yet captured; SVG or sketch placeholder until then.

• **Captured** — file path replaces this entry; image embedded in `site/index.html`.

• **Reshoot** — captured but rejected on quality / framing; queued.

Photography conventions

• **Backdrop:** kraft butcher paper or off-white seamless for hero shots; raw shop bench OK for build-process shots.

• **Lighting:** one diffused softbox 45° camera-left for hero/detail; ambient shop light + tripod-stable shutter for process.

• **Format:** all photographs landscape 3:2 unless explicitly portrait. Hero is portrait 4:5.

• **Resolution:** 4000 × 6000 minimum at capture; export 2000 × 3000 web JPEG for the site.

• **Naming:** `images/<SHOT_ID>_<short-slug>.jpg` (e.g., `images/HERO-001_v1-std-finished.jpg`).

risks.md

Project artifact.

Risks — Wood Shell Tongue Drum

Red-team pass over the design. Each risk has a category, severity, mitigation, and an attached verification test that is mirrored into `validation.csv` so the empirical-learning loop closes around it once the prototype exists.

This is a *new* design — no commercial wood-shell tongue drums exist in the round-body × dome-top × premium-tonewood format at this depth — so the risk surface is broader than for a refinement of an established pattern. The first physical prototype's measurements will retire most of the acoustic risk; structural and ergonomic risks need pre-prototype mitigation.

Acoustic

A1 — Curved-cantilever +2.5 % multiplier is unmeasured for our specific dome rise (V2, V4)

Severity: high (V2, V4 only)

Description: The tongue length correction `L_curved = L_flat × 1.025` is a starting estimate from cylinder-shell theory for shallow domes. The actual factor for our specific dome rise (0.75″ on Standard, 4.7 % of OD) is not measured. If the actual factor is, say, 1.04, every tongue cut at 1.025-multiplier length will sit ~30 cents flat across the field.

Mitigation: before cutting all 11 tongues on a V2/V4 build, cut ONE tongue at the predicted length, measure pitch with a Korg OT-120, back-solve the actual multiplier (`m_actual = sqrt(f_meas / f_predicted)` adjusted for the cut geometry), and apply the corrected multiplier to the remaining 10. The first dome build is also the first measurement entry into the per-family corrections database.

Verification test: "first-tongue calibration" entry in `validation.csv` for V2/V4 only — measured/predicted ratio with acceptance band 0.95–1.10.

A2 — Helmholtz under-coupling on V2 and V4 with preset gu port

Severity: medium

Description: V2 and V4 ratios on Standard 16″ are 0.61 and 0.73 respectively — under-coupled. The cavity's Helmholtz resonance sits well below the ding and won't reinforce it. The drum will sound fine but not gain the bass extension the design promises.

Mitigation: the gu port preset (2.5″ on Standard) is a reviewable starting point, not a fixed feature. For V2/V4 builds, start with a pilot hole, step up in 0.25″ increments, and expect a larger effective port than V1 because larger port area raises `f_H`. The Standard 16″ V2 lower coupled edge is estimated near 3.3″; if that is structurally or aesthetically too large, switch to a port sleeve / neck-length experiment and record the revised model.

Verification test: Helmholtz cavity row in `validation.csv` per variant; pass = `f_H/f_ding` ∈ [0.80, 1.20].

A3 — Floor Pouf D3 ding tongue exceeds radial fit cap

Severity: high (Floor Pouf only)

Description: Computed flat-cantilever D3 ding length on 0.5″ Padauk = 9.122″. Fit cap (radius – 1″ clearance) = 9.0″. Tongue overhangs by 0.12″ — does not fit.

Mitigation: three options, ordered by preference:

1. **Cedar substitute** for the Floor Pouf soundboard (K = 29,013). Computed length 8.34″ — fits with 0.66″ margin. Tonal trade: brighter, more guitar-like, less bass.

2. **Thin the soundboard** from 0.5″ to 0.4375″. Computed length 8.53″ — fits with 0.47″ margin. Trade: stiffness across a 20″ span drops; risk of soundboard sag.

3. **Raise ding to D■3** (one semitone, MIDI 51, 155.6 Hz). Computed length 8.86″ at 0.5″ Padauk — fits with 0.14″ margin. Trade: scale shift; player must learn the new key.

Verification test: ding tongue length measurement vs cap before slit cuts; abort cut if margin < 0.10″.

A4 — Soundboard rabbet leaks → Helmholtz Q drops

Severity: medium

Description: Helmholtz resonators need an airtight cavity to sustain bass. A leaky rabbet joint will let the cavity bleed pressure on each tongue strike, killing the bass extension.

Mitigation: lathe-trued rim (± 0.005″ TIR), slip-fit rabbet (0.005″ clearance), uniform Titebond III bead, two-caul clamp. Smoke or vacuum test before finishing.

Verification test: "bowl-top airtight seal" row in `validation.csv` — pass/fail vacuum or smoke test post-glue, pre-finish.

A5 — End-correction error in Helmholtz formula

Severity: low

Description: The Helmholtz formula uses `L_neck = soundboard_thickness` for flat tops. The actual oscillating air column extends slightly above and below the soundboard plane (the "end correction" of an open pipe). For tongue-drum kerfs (slit, not circular) this is small but nonzero.

Mitigation: the design currently absorbs the error in the 0.88-ratio safety margin (so even if the actual `f_H` runs 5–10 % low, it stays in the coupled band). If first-prototype `f_H` measures > 10 % below predicted, add an empirical end-correction term to the workbook formula and update the per-family corrections database.

Verification test: "Helmholtz cavity" row in `validation.csv`; if `(predicted - measured) / predicted > 0.10`, file an empirical-correction update.

A6 — A4 ≠ 440 Hz on the prototype

Severity: low (failure mode, not a design risk — but the catch is structural)

Description: Tongue 4 on the recommended prototype is A4 (440 Hz). If the measured A4 is off > 5 cents, the K-constant of our specific Padauk stock differs from the library value (24,438). All other tongue lengths will be proportionally wrong.

Mitigation: cut tongue 4 first as a calibration tongue; measure A4; back-compute `K_actual = f_measured × L² / t`; update the workbook K-cell; recompute all 10 remaining tongue lengths before cutting.

Verification test: "A4 = 440 Hz sanity" row in `validation.csv`.

Structural

S1 — Soundboard slit-area exceeds 50 % limit causes rim-zone failure

Severity: medium

Description: Removing too much material from the soundboard (slits + perimeter rabbet relief) leaves the rim zone too weak; under cantilever loading from the tongues plus rim constraint from the rabbet, the rim could chip or split, especially in Padauk's straight-grain regions.

Mitigation: total slit kerf + rabbet area held at ~30 % of soundboard area on V1 Standard (verified against the 11-tongue layout). Stay-zones at least 0.75″ wide between adjacent tongue slits.

Verification test: measure slit-area fraction during CAD/CNC programming; reject layouts > 45 %.

S2 — Stave glue-line failure under thermal/humidity cycling

Severity: low-medium

Description: Black Walnut staves have axial grain and tangential growth. Seasonal humidity cycling could open glue lines (especially if the shell finishes too soon after glue-up).

Mitigation: Titebond III is rated 4000 psi shear (waterproof PVA), well above structural minimum. Cure full 24 h before lathe-truing. Acclimate stock 2 weeks pre-glue. Apply finish to *exterior* only — leave the interior unfinished so the wood can equalize humidity through one face.

Verification test: glue-line shear sample test (one sacrificial joint per glue batch); reject batch if test joint fails < 3000 psi.

S3 — Soundboard cup or warp during finish drying

Severity: medium

Description: Finishing only one face of a soundboard (Watco on top, unfinished underside) creates a moisture gradient. If finish is applied too thick or too fast, the disc cups, lifting tongue tips off neutral and detuning the field.

Mitigation: thin coats (3 light coats vs 1 heavy), 24 h between coats, finish in same humidity environment as glue-up. Hold the disc flat on cauls for the first cure.

Verification test: post-finish flatness check across diagonal — disc should be flat within 1/64″ over 16″ span.

S4 — Bowl ring-stack delamination at lathe truing (V3, V4)

Severity: medium (V3, V4)

Description: A segmented bowl with 8+ rings stacked has many glue joints under shear during lathe truing. A weak joint shows up as a "spinning piece flying off" — both a build hazard and a quality failure.

Mitigation: glue each ring closed BEFORE stacking; cure 24 h between layers; verify each ring's closure with a feeler gauge (< 0.005″ at glue lines) before adding the next. Lathe at moderate speed (800–1200 rpm) with sharp tools to minimize cutting forces.

Verification test: feeler-gauge check per glue line during stack-up; reject ring if any joint > 0.005″.

Ergonomic / playability

E1 — Standard 16″ too heavy for lap play

Severity: low

Description: Estimated weight 6–7 lb for V1 Standard 16″ — heavier than a lap-played Hapi steel tongue drum (~4 lb). Player fatigue on extended sessions.

Mitigation: acceptable trade for the wood acoustic benefit. Optional carry handle (`HANDLE-ROPE-S` in BOM) makes the drum portable. Floor Pouf 20″ is explicitly intended for floor placement, not lap.

Verification test: finished-drum weight measurement; document in catalog row but don't gate on this.

E2 — Tongue layout reach for 5th-percentile player hand

Severity: low

Description: Tongues at radius 6.5″ from ding centerline require a ~13″ hand-span reach (across the ding). A 5th-percentile adult female hand-span is ~7″, so the maximum-radius tongues are at the edge of comfortable reach.

Mitigation: the radial coords are estimates and will be tightened during SolidWorks parametric placement against the actual shell radius. Recommended max radius from ding centerline: 6.0″ on Standard 16″, with the 7-tongue ergonomic envelope inside 5.0″.

Verification test: ergonomic photo with two-hand reach over the finished drum; note any tongue requiring a player to lean to strike.

Supply / material

SUP1 — Padauk 3/8″ thickness availability

Severity: low

Description: Padauk is commonly stocked in 4/4 (rough 1″, S2S 13/16″) but 3/8″ is non-standard. Suppliers may resaw to spec but with no guarantee of edge straightness or grain orientation match.

Mitigation: order 4/4 stock and resaw + plane to 3/8″ in-shop. Adds 1 h labor per soundboard but guarantees control over thickness uniformity.

Verification test: confirm 3/8″ stock at order time; if not available, switch to 4/4 + resaw plan.

SUP2 — Mule hide / drum-skin substitution attempts

Severity: low (filed for awareness)

Description: A reviewer or supplier might confuse this design with a membrane drum (conga, ashiko) and try to ship drum heads. This is a slit-tongue idiophone — no skins.

Mitigation: explicit "N/A — no skins required" in the supplier RFQ and BOM.

Fit / finish

F1 — Slit edges chipped during sanding

Severity: medium

Description: A chipped slit edge changes the tongue's effective length and shifts pitch by tens of cents. Sanding past 320 grit near the slits is risky.

Mitigation: sand the soundboard *before* CNC slit routing for any work that requires aggressive grits. Post-slit, hand-sand only with light pressure, 320–600 grit, parallel to slit length, never across.

Verification test: visual inspection under raking light; reject any chip > 0.020″.

F2 — Finish bleeds into slits and stiffens tongue base

Severity: low-medium

Description: Watco Danish Oil applied to the soundboard top can wick into the slit kerfs and accumulate at the tongue-base radius, locally stiffening the cantilever and shifting pitch ~5–15 cents sharp per tongue.

Mitigation: apply finish *before* slit cutting if the finish workflow allows. Otherwise, use a thinned finish (1:1 with mineral spirits) on the soundboard top, wipe excess immediately, and never flood the slits.

Verification test: post-finish A4 measurement; if A4 has shifted > 5 cents from pre-finish, document as a finish-bleed effect and quantify for the next build.

Risk summary table

ID	Category	Severity	Variants affected	Mitigation in place	Verification gate
A1	Acoustic	High	V2, V4	Yes (first-tongue calib)	first-tongue cal in validation.csv
A2	Acoustic	Medium	V2, V4	Yes (larger effective port or port-sleeve model)	f_H/f_ding ratio band
A3	Acoustic	High	Floor Pouf only	Yes (Cedar / D■3)	ding length vs cap
A4	Acoustic	Medium	All	Yes (rim TIR + smoke test)	bowl-top airtight seal
A5	Acoustic	Low	All	Yes (safety margin)	f_H deviation > 10 %
A6	Acoustic	Low	All	Yes (calibration tongue)	A4 sanity row
S1	Structural	Medium	All	Yes (slit-area cap)	slit-area fraction ≤ 45 %
S2	Structural	Low-Med	All (stave path)	Yes (Titebond III)	glue-line sample shear test
S3	Structural	Medium	All	Yes (thin coats)	post-finish flatness
S4	Structural	Medium	V3, V4	Yes (per-ring cure)	feeler gauge per joint
E1	Ergonomic	Low	All	Optional handle	weight measurement (no gate)

E2	Ergonomic	Low	All	Yes (radius cap)	reach photo on prototype
SUP1	Supply	Low	All	Yes (resaw plan)	order-time confirm
SUP2	Supply	Low	All	Yes (RFQ note)	N/A
F1	Fit/Finish	Medium	All	Yes (sand sequence)	edge inspection
F2	Fit/Finish	Low-Med	All	Yes (thin finish)	A4 shift post-finish

---

Highest-risk unknowns (to be retired by Phase 1 measurements)

1. **Curved-cantilever multiplier for our specific dome rise.** Will be measured on the first V2 build; until then, V2/V4 cuts are conditional on calibration.

2. **Helmholtz end-correction term.** Will be measured on the first V1 build; the design currently absorbs the uncertainty in the safety margin but the workbook formula may need an empirical update.

3. **Padauk K-constant for our specific stock.** The library value 24,438 is from textbook E and ρ; the actual stock may vary 2–5 %. A4 calibration on Phase 1 catches this.

4. **Bowl-top joint long-term durability.** Rabbet + Titebond III is the working assumption; 12+ months of seasonal cycling will reveal whether the joint holds.

5. **Player ergonomics on Floor Pouf.** A 20″-OD drum with the largest tongue at 9″+ length is a new size class for Tony's catalog. First-prototype reach photos will inform whether the radial layout needs revision.

validation-report.md

Project artifact.

Validation Report — Wood Shell Tongue Drum

Generated during the v4 public-readiness pass on 2026-05-08.

Clean Checks

Gate	Result
Required Mode A files present	Pass
README is instrument-specific and includes hero drawing	Pass
Parametric design source exists	Pass (`wood-shell-tongue-drum-design-table.xlsx`)
BOM, sourcing, cut list, validation CSVs present	Pass
Assembly, RFQ, visual BOM, photo shot list present	Pass
Risk register exists with acoustic, structural, ergonomic, supply, and finish risks	Pass
Build-log site exists	Pass (`site/index.html`)
Print packet PDF exists	Pass
Capstone markdown exists	Pass

Fixed In This Pass

Finding	Files updated	Resolution
Gu-port tuning direction was reversed in several notes	`design.md`, `README.md`, `assembly-manual.md`, `risks.md`, `validation.csv`, `site/index.html`	Clarified that increasing port area raises Helmholtz frequency; under-coupled V2/V4 variants need a larger effective port or revised neck/cavity model
CNC/laser/jig strategy was present but scattered	`cnc/README.md`, `cnc/jig-and-template-plan.md`	Added a single fixture/template decision matrix for public review
README status undersold generated packet outputs	`README.md`	Marked capstone markdown and printable packet outputs as complete; noted that PPTX refresh needs python-pptx

Escalated / Human-Owned

Gate	Remaining action

| Physical tuning validation | Build V1 Standard 16 in prototype; fill `measured_value`, `cents_error`, and environment fields in `validation.csv` |

| Supplier facts | Re-check current price, stock, shipping, and substitutes before buying |

| Capstone PPTX refresh | Existing `capstone-deck.pptx` is retained, but the local generator did not refresh PPTX because python-pptx is unavailable in this environment |

| CAD and G-code | Generate only after Phase 1 measurements calibrate Padauk K, rim seal behavior, and Helmholtz end correction |

| Real photos/audio | Replace concept/SVG placeholders with shop process photos, final instrument photos, and tuning recordings after build |